

Bankruptcy Codes and Innovation

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We argue that when bankruptcy code is creditor friendly, excessive liquidations cause levered firms to shun innovation, whereas by promoting continuation upon failure, a debtor-friendly code induces greater innovation. We provide empirical support for this claim by employing patents as a proxy for innovation. Using time-series changes within a country and cross-country variation in creditor rights, we confirm that a creditor-friendly code leads to a lower absolute level of innovation by firms, as well as relatively lower innovation by firms in technologically innovative industries. When creditor rights are stronger, technologically innovative industries employ relatively less leverage and grow disproportionately slower. (JEL G3, K2, O3, O4, O5)

Existing empirical evidence indicates that legal institutions of an economy affect its financial organization and economic growth.¹ Less well understood, especially empirically, is whether legal institutions that govern financial contracts affect the nature of real investments in an economy. In this paper, we focus on one specific aspect of this overarching theme: Does the nature of bankruptcy code affect the extent of innovation in an economy?²

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¹ For example, the literature has documented an effect of legal institutions on the nature of external financing of enterprises (La Porta et al. 1997, 1998), the ownership structure of firms (La Porta, Lopez-de-Silanes, and Shleifer 1999), the mix between market- and bank-dominated finance (Allen and Gale 2000), and economic growth through the provision of financial access to firms (King and Levine 1993; Rajan and Zingales 1998).

² Entrepreneurs certainly seem to believe there is such a link: "The wider lesson is not to stigmatize failure but to tolerate it and learn from it: Europe's inability to create a rival to Silicon Valley owes much to its *tougher bankruptcy laws*." Lessons from Apple, *The Economist*, June 7, 2007 (emphasis added).

The effect of bankruptcy codes on real investments can be understood in the context of conflicts of interest arising between firmowners/equityholders and creditors at time of distress, as studied theoretically in a large body of literature starting with Gertner and Scharfstein (1991).³ Consider two polar cases of the bankruptcy code: First, the debtor-friendly code where equityholders retain all control rights in bankruptcy; and second, the creditor-friendly code where all control rights are transferred to a firm's creditors. Also consider two technologies: innovative and conservative. In financial distress, the firm's existing investment may be inefficiently liquidated under the creditor-friendly code, whereas it may be inefficiently continued under the debtor-friendly code. While this trade-off arises for both technologies, greater risk inherent in the innovative technology accentuates the deadweight costs arising from liquidation under the creditor-friendly code but mitigates the deadweight costs from continuation under the debtor-friendly code. Thus, a creditor-friendly code discourages ex ante risk taking and innovation relative to a debtor-friendly code.

Firms' investment decisions are also affected by their choice of the optimal financing mix. When the code becomes more creditor friendly, firms unwind the inefficiencies arising from bankruptcy codes by optimally financing the innovative technology with disproportionately lower debt when compared to the conservative technology. However, there is a cost to lowering debt (for example, the foregoing of tax shields or incentive benefits related to debt). Therefore, compared to the debtor-friendly code, under the creditor-friendly code, the firm's value associated with the innovative technology is disproportionately lower than that associated with the conservative technology. We thus predict that aggregate innovation would be lower in countries where the bankruptcy code is more creditor friendly. The mechanism for this effect yields the following difference-in-difference prediction: when the bankruptcy code becomes more creditor friendly, the value of an optimally financed innovative firm reduces disproportionately more than that of an optimally financed conservative firm.

We provide evidence supporting this effect of bankruptcy codes on innovation using patents issued by the U.S. Patent Office (USPTO) to U.S. and foreign firms from 1978 to 2002, and citations to these patents, as constructed by Hall, Jaffe, and Trajtenberg (2001). The "industry" level classification we employ pertains to the patent classes in these data. We measure innovation for an industry in a given year by the number of patents applied for in that year (and subsequently granted) in that industry, the number of all subsequent citations to these patents, and the number of firms filing for patents in that industry.⁴ We

³ A formal model of arguments to follow can be found in Acharya and Subramanian (2009), a theoretical addendum to the current paper.

⁴ We discuss in detail, in the main body of the paper, the relative merits of different patent-based measures of innovation, and indeed of patents themselves as a measure of innovation.

use Djankov, McLiesh, and Shleifer (2007) for information on a country-level creditor rights index⁵ and its within-country change.

We follow a twofold empirical strategy. First, since creditor rights are likely to be correlated with other country-level unobserved factors, we exploit country-level exogenous changes in creditor rights to conduct difference-in-difference tests of the causal effect of creditor rights on innovation. Second, to highlight the causal mechanism, we test in the pooled cross-section whether a higher creditor rights index for a country is associated with relatively lower innovation in industries that have a higher propensity to innovate. We also use the exogenous creditor rights changes to confirm the relative effect obtained in the cross-sectional tests. To conduct these cross-sectional and time-series tests of the causal mechanism, we follow the methodology introduced by Rajan and Zingales (1998) and rank patent classes by their patenting intensity in the United States.⁶ Except for this ranking of industries, we exclude the patents filed by the U.S. firms from all other analyses.

In our difference-in-difference tests, we find strong evidence that strengthening of creditor rights lowers innovation. Compared to the “control” group of counties that did not undergo a creditor rights change, the “treatment” group of countries that underwent a creditor rights increase (decrease) generated 9.7% less (10.7% more) patents, 13.3% less (15.4% more) citations to these patents, and 8.4% less (9.2% more) patenting firms. In estimating this effect, we include country- and time-fixed effects that are required to identify the difference-in-difference, as well as fixed effects for each patent class to control for industry-level unobserved factors. We also control for (i) a country’s bilateral trade with the United States in each of its industries using its exports and imports with the United States in different years; (ii) a measure of the country’s comparative advantage in an industry using the ratio of value added of an industry in a country in a given year to the total value added for the country that year; and (iii) the GDP per capita of the country. To control for the effect of changes in patent protection in some countries during our sample period, we reestimate the difference-in-difference by excluding these countries and find the effect to be similar. To alleviate concerns about reverse causality, we investigate the dynamic effect of the creditor rights changes and find that the effect manifests only two years after the change.⁷ Furthermore, the effect of the creditor rights

⁵ This is a score between 0 and 4, based on a score of one each for whether there is no automatic stay on secured creditors’ rights, secured creditors are paid first in bankruptcy, no majority creditor consent is required for reorganization, and management does not stay in place upon bankruptcy.

⁶ As Cohen, Nelson, and Walsh (1996) find in their survey of patenting across various U.S. industries, the propensity to patent is largely driven by technological characteristics of an industry. Since the United States has had the best financial markets necessary to fund financially constrained high-growth sectors (Rajan and Zingales 1998), the most vibrant research environment, the world’s most advanced patenting system (Kortum and Lerner 1999), and the preferred patenting jurisdiction for innovating firms internationally (Cantwell and Hodson 1991), we can make the reasonable assumption that the U.S. ranking best reflects the technological propensity of an industry to innovate.

⁷ Furthermore, based on existing studies of what caused the creditor rights to change, we argue that in the case of some countries, the change occurred for exogenous reasons (e.g., to promote employment or protect domestic

change is experienced both at the intensive and extensive margins since we find similar effects using the median-, average-, and citation-weighted average number of patents.

Next, we undertake cross-sectional tests to highlight the causal mechanism for the effect of creditor rights on innovation. Since our explanatory variable here is the interaction between country-level creditor rights and the industry-level innovation intensity, we include country- and industry-fixed effects together with year-fixed effects in all our specifications. We find that the coefficient of the interaction between creditor rights and innovation intensity is uniformly negative and statistically and economically significant. We show that these cross-sectional results are also robust to time-varying heterogeneity at the country level (using country by time-fixed effects) and industry level (using industry-by-time-fixed effects), as well as to effects of a country's bilateral trade with the United States, its industry-level comparative advantages, and its GDP per capita.

Our most compelling evidence on the causal mechanism comes from the time-series tests. Here, we conduct a third-difference test by computing a difference-in-difference estimate for the interaction of creditor rights with innovation intensity. We find that while the difference in innovation between two adjacent industries is statistically indistinguishable in countries that did not experience a creditor rights change, in countries that underwent an increase (a decrease) in creditor rights, the more innovative industry generated 10.3% less (11.5% more) patents, 56.4% less (29.3% more) citations to these patents, and 9.5% less (10.5% more) patenting firms than its adjacent less-innovative industry.

Having confirmed the effect of bankruptcy codes on innovative activity, we examine the effect of bankruptcy codes on leverage choice. Due to limits on data availability, we focus on the G-7 countries and study the relationship between leverage and creditor rights. We find that when creditor rights are stronger, innovative industries take on relatively less leverage (book debt, market debt, debt inclusive of all nonequity liabilities, and debt net of cash and cash equivalents) compared to other industries. Thus, firms in innovative industries do appear to unwind the effect of stronger creditor rights by undertaking smaller quantities of debt and keeping more cash reserves.

Finally, we ask how the differential impact of bankruptcy code on innovative versus noninnovative industries impacts the growth rates of these industries. In regressions using the growth rates for each ISIC industry in a country, we find the coefficient of the interaction between creditor rights and industry-level innovation intensity to be strongly negative. This effect is also economically significant and robust to including the Rajan and Zingales (1998) measures of financial development and their interaction with external financial dependence.

industries); in others, the change occurred precisely to promote innovation and give managers-entrepreneurs a fresh start in default; whereas in some others, the change was part of an overall package of reforms designed to stimulate growth following recessions.

Note that this evidence does not rely on any data on patents other than the U.S. data, which are employed to calculate the intrinsic innovation intensity of industries. This evidence is thus immune to concerns that patents may be an imperfect proxy for all innovation activity.

The paper proceeds as follows. Section 1 presents our empirical evidence. Section 2 discusses our innovation proxy and related literature. Section 3 concludes.

1. Empirical Evidence

1.1 Empirical strategy

We investigate whether a more creditor-friendly bankruptcy code lowers innovation in a country. To understand our test designs, consider two industries in two countries: biotechnology and textiles in the United States and Germany. Firms in the biotechnology sector have a higher propensity to innovate and have riskier cash flows than firms in the apparel industry, while the German bankruptcy code is more creditor friendly than the U.S. bankruptcy code (Djankov, McLiesh, and Shleifer 2007). We investigate whether innovation in Germany is lower than that in the United States. For this purpose, we follow a twofold empirical strategy.

First, since creditor rights are expected to be largely collinear with other country-level unobserved factors, we exploit country-level exogenous changes in creditor rights to conduct difference-in-difference tests: the difference in the level of innovation in a country before and after the creditor rights change, compared to the same difference for countries that did not experience a creditor rights change. The difference-in-difference test provides direct evidence that stronger creditor rights lead to lower innovation.

Second, we focus on the mechanism that leads to this effect. We predict that the value from innovation for biotechnology firms minus that for apparel and textile firms is expected to be higher in the United States than in Germany. Figure 1 illustrates this interaction effect. In this figure, we plot across time the ratio of realized number of patents and citations for two innovation-intensive industries (biotechnology and surgery and medical instruments) relative to a benchmark conservative industry (apparel and textile) for the United States vis-à-vis Germany (the top two plots) and the United States vis-à-vis Japan (the bottom two plots). Note that Japan, like Germany, also has a higher creditor rights index than the United States. In each of the two innovation-intensive industries, the ratio for the United States is substantially higher than that for either Germany or Japan, often by a factor of five. The factor increases in most cases over time right from 1978 (the beginning of our data and, in fact, the year the United States passed the Bankruptcy Reform Act, making its code even more debtor friendly). In the econometric variant of this visual test, we examine the interaction of a country-level index of creditor rights with

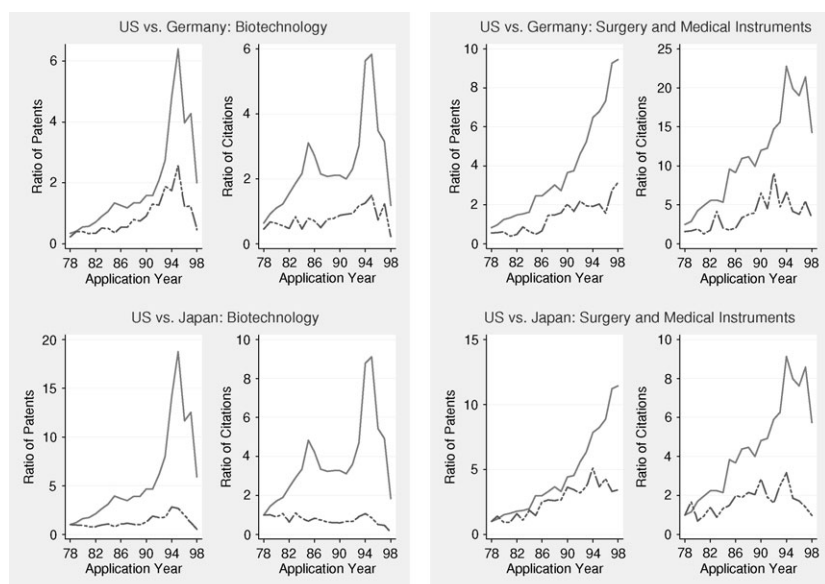


Figure 1
Differences in innovation between innovation-intensive and nonintensive industries for the United States vis-à-vis Germany and Japan

This figure plots the time series of the ratio of patents and ratio of citations in two innovation-intensive sectors (biotechnology and surgery and medical instruments) to those in a nonintensive sector (textiles and apparel) for the United States vis-à-vis Germany and Japan. The solid line shows the trend for the United States, while the dotted line shows the same for Germany/Japan.

proxies for the innovation intensity of an industry. Here, we include country- and industry-level fixed effects to control for the time-invariant unobserved heterogeneity at these levels.

We also use the creditor rights changes to reexamine the interaction effect described above. Here, we estimate a difference-in-difference for the interaction term, thus providing a third-difference test for our hypothesis.

1.2 Data and proxies

To implement the cross-sectional and third-difference time-series tests described above, we construct a proxy for an industry's intrinsic propensity to innovate. We describe the construction of this proxy after describing our data and sample and our proxies for innovation.

To proxy for innovation, we use patents filed with the USPTO and the citations to these patents, compiled in the National Bureau of Economic Research (NBER) Patents File (Hall, Jaffe, and Trajtenberg 2001). The NBER patent data set provides, among other items, annual information on patent assignee names, the number of patents, the number of citations received by each patent, the technology class of the patent, and the year in which the patent application is filed. The data set covers all patents filed with the USPTO by firms from around

85 countries. We exploit the technological dimension of the data generated by “patent classes.” Over the years, the USPTO has developed a highly elaborate classification system for the technologies to which the patented inventions belong, consisting of about 400 patent classes. During the patent examination process, patents are assigned to detailed technologies as defined by the patent class. The USPTO performs these assignments with care to facilitate future searches of the prior work in a specific area of technology (Kortum and Lerner 1999).

We date our patents according to the year in which they were applied for. This avoids anomalies that may be created due to lag between the date of application and the date of granting of the patent (Hall, Jaffe, and Trajtenberg 2001). Note that although we use the application year as the relevant year for our analysis, the patents appear in the database only after they are granted. Hence, we use the patents actually granted (rather than the patent applications) for our analysis.⁸

1.2.1 Proxies for innovation. We use three broad metrics to measure innovation. The first is a simple patent count of the number of patents that were filed in a particular year in a specific patent class. As our second metric of innovative activity, we use the citations that are made to the patents in a specific patent class. Citations capture the importance and drastic nature of innovation. This proxy is motivated by the recognition that the simple count of patents does not distinguish breakthrough innovations from less significant or incremental technological discoveries.⁹ Intuitively, the rationale behind using patent citations to identify important innovations is that if firms are willing to further invest in a project that is building upon a previous patent, it implies that the cited patent is influential and economically significant. In addition, patent citations tend to arrive over time, suggesting that the importance of a patent may be revealed over a period of time and may be difficult to evaluate at the time the innovation occurred. Finally, citations help control for country-level differences arising in the number of patents due to differences in the number and size of firms.

As our third measure of innovative activity, we employ the number of patenting firms in a patent class. The USPTO defines “assignee” as the entity to which a patent is assigned. A simple count of the number of assignees in a patent class in a given application year provides a measure of the number of patenting entities. Since patents may be assigned to individuals, corporations, or academic departments, the USPTO classifies patents according to the type

⁸ A caveat about potential biases created by the use of application year, particularly in the case of foreign patents, is in order. Since foreign firms usually file patents with the domestic patent office and then with the USPTO, readers may believe that the application year recorded with the USPTO does not capture the exact timing of the innovation. However, the Paris Convention, which governs such firms filing in both the domestic and foreign countries, mandates that if the inventor files a foreign patent application in any other Paris Convention signatory state within twelve months of the domestic filing, overseas patent-granting authorities will treat the application as if it were filed on the first filing date. Therefore, the application year recorded with the USPTO would coincide with the application year for the domestic patent of the foreign firm.

⁹ Pakes and Shankerman (1984) show that the distribution of the importance of patents is extremely skewed, i.e., most of the value is concentrated in a small number of patents. Hall, Jaffe, and Trajtenberg (2005), among others, demonstrate that patent citations are a good measure of the value of innovations.

of assignees using the “assignee code.” Assignee codes equal to 2 and 3 identify U.S. non-government organizations (mostly corporations) and non-U.S., non-government organizations (mostly corporations), respectively. These categories account, respectively, for 47.2% and 31.2% of all patents filed with the USPTO.

While the number of patents, the number of citations, and the number of patenting firms represent measures of aggregate innovation, we also examine average measures of innovation in a patent class. We employ the median number of patents, the simple average number of patents, and the citation-weighted average number of patents applied in a patent class. Employing the aggregate measures of innovation, together with these average measures, enables us to examine both the extensive and intensive margins of innovative activity.

Patents have long been used as an indicator of innovative activity in both micro- and macro-economic studies (Griliches 1990). Although patents provide an imperfect measure of innovation, there is no other widely accepted method that can be applied to capture technological advances.¹⁰ Nevertheless, we are aware that using patents has its drawbacks. Not all firms patent their innovations, because some inventions do not meet the patentability criteria and because the inventor might rely on secrecy or other means to protect its innovation. In addition, patents measure only successful innovations. To that extent, our results are subject to the same criticisms as previous studies that use patents to measure innovation (e.g., Griliches 1990; Kortum and Lerner 1999).

1.2.2 Proxy for innovation intensity. The “industry” level classification we employ pertains to the patent classes in the data. We identify an industry’s (patent class’s) propensity to innovate by the number of patents filed by the median U.S. firm in that industry. This corresponds to using the patents under USPTO assignee code “2.”

The choice of the United States for the ranking of industries is for several reasons. First, the United States has the most populated data across various patent classes and over time. Second, it had the most well-developed financial markets over our sample period, and these have been shown to be necessary for funding of constrained but high-growth sectors (Rajan and Zingales 1998). Last but not least, the United States has had the most vibrant research environment in universities and the most open immigration policy for enrolling scholars in these universities. While using the patents filed by U.S. firms is the most natural choice for proxying innovation intensity, as a robustness check, we also employ the USPTO patents filed by Japanese firms to generate a proxy for innovation intensity.

¹⁰ As an alternative to patents as proxies of innovation, R&D spending across different industries could be a potential proxy for innovation intensity. However, in a cross-country setting, this presents several challenges. For example, accounting norms, particularly whether R&D is capitalized or expensed, would have a mechanical effect on R&D spending. Griliches (1990) emphasized that there is a strong relationship in the United States between R&D and the number of patents received at the cross-sectional level, across firms and industries. The median *R*-squared is of the order of 0.9.

We then make the assumption that the propensity to innovate is driven primarily by the technological characteristics of firms in that industry. This assumption can be justified along the following grounds. First, Cohen, Nelson, and Walsh (1996) find in their survey of patenting across various U.S. industries that the propensity to patent is largely driven by technological characteristics of an industry. Hence, we reason that these technological characteristics carry over to other countries. For example, the correlation in patent class ranking between the United States and Germany (Japan) is 0.676 (0.618), on average in the time-series, its value ranging from a minimum of 0.63 (0.54) to a maximum of 0.76 (0.71).

Second, Hall, Jaffe, and Trajtenberg (2001) find that the number of patents filed per year and the number of citations per year to these patents vary with the industry category, with computers and communications having the highest number of patents filed per year and the highest number of citations to these patents per year, and the more mature mechanical industry exhibiting the lowest values on both counts. They describe the reasons for these differences as primarily technological. Cohen, Nelson, and Walsh (1996) contrast between discrete and complex technologies and argue that firms file more patents and cite each other's patent more when the technology is "complex."

While Rajan and Zingales (1998) construct a time-invariant industry-level proxy for external financial dependence, our proxy for innovation intensity is time varying. Hall, Jaffe, and Trajtenberg (2001) show, in Figure 5 of their study, that there have been changes in the share of patents occupied by the different industry categories over the period 1978–2002. This is an important observation suggesting that the technological innovation intensity of industries has changed in a relative fashion over time. This point has also been stressed by Kortum and Lerner (1999), who find, for example, that the number of patents filed in biotechnology and software industries has risen considerably since the late 1970s, both absolutely and as a share of total patenting in the United States. Kortum and Lerner (1999) suggest that the principal reason for this pattern is the intertemporal variation in the level of innovation across different industries, particularly due to the arrival of technological shocks.

1.2.3 Creditor/debtor friendliness of the bankruptcy code. We view a bankruptcy code to be more creditor friendly if the code provides more rights to creditors in bankruptcy. Our data on the cross-section and time series of country-level index of creditor rights come from Djankov, McLiesh, and Shleifer (2007). This is a score between 0 and 4, based on a score of one each for whether there is no automatic stay on secured creditors' rights, secured creditors are paid first in bankruptcy, no majority creditor consent is required for reorganization, and management does not stay in place upon bankruptcy. The creditor rights score is simply an index and the extreme scores of 0 and 4 do not literally correspond to a lack of any creditor rights or debtor rights. They simply indicate that bankruptcy codes confer control rights relatively more on one set of a firm's

claimants, likely favoring the bargaining and reorganization in their favor. In particular, a creditor right index of 0 for a country does not imply that there is no supply of debt financing in this country.

Note that although the patent data are available from 1963 onward, the information on the country-level index of creditor rights provided by Djankov, McLiesh, and Shleifer (2007) starts only in 1978. Therefore, the time period of our sample is 1978–2002.

1.2.4 Summary statistics. Table 1 lists the 85 countries that appear in the NBER patents data file along with the aggregate number of patents filed by firms in the country, the aggregate number of citations received by these patents, the number of firms filing patents from this country, and the mean and median numbers of patents filed by firms in this country. The table also displays the index of creditor rights in each of these countries. While for most countries the creditor rights index is unchanged over the sample period, a few countries experienced a change in creditor rights. For these countries, the values of the creditor rights are listed sequentially.¹¹

1.3 Difference-in-difference test using changes in creditor rights

To provide causal evidence on the impact of creditor rights on innovation, we exploit country-level exogenous changes in creditor rights. A total of twelve countries underwent a change in creditor rights index over our sample period. Seven of these (Canada, Finland, Indonesia, Ireland, Israel, India, and Sweden) experienced a decrease in creditor rights by one, and five (Denmark, United Kingdom, Lithuania, Romania, and Russian Federation) experienced an increase. Table 2 lists these countries and their year of change in creditor rights. Since creditor rights changed in some countries in our sample, but not in others, this “natural experiment” offers the opportunity to estimate the causal effect of creditor rights change through a difference-in-difference test.

To understand how this difference-in-difference is calculated, consider the effect on innovation due to the creditor rights change in Israel in 1995. A naïve estimate would be to simply compute the difference in innovation before and after the creditor rights change. However, this estimate would also be affected by time trends that coincide with the creditor rights change, as well as other economy-wide factors. To control for such factors, we also estimate this difference in innovation for a country such as Germany, which did not undergo a creditor rights change in 1995. The difference estimated for Germany provides an answer to the counterfactual question: “What would have been the difference in innovation in Israel if the creditor rights change had not occurred?” The

¹¹ Readers may query how we treat the patents that are filed by U.S. subsidiaries of foreign firms, and whether the inclusion or exclusion of such patents affects our results. We identify such patents as those where the country of the “assignee” is non-U.S., but the country of the “inventor” is recorded as the United States. Of the 3,333,701 patents in our sample, we identify 21,489 patents (0.6%) issued to U.S. subsidiaries of foreign companies. Not surprisingly, excluding these patents does not change our results.

Table 1
Summary statistics by country

Country	Number of patents	Number citations	Number firms	Mean patents	Mean citations	Creditor rights
Argentina	137	318	130	1.2	2.7	1
Australia	6,452	26,689	5,448	2.1	8.8	3
Austria	6,134	20,946	4,430	2.2	7.4	3
Belgium	6,168	22,641	3,769	2.7	9.8	2
Brazil	648	1,699	554	1.2	3.3	1
Bulgaria	253	972	239	1.2	4.5	1,2
Canada	24,141	124,569	18,583	4.2	21.9	2,1
Chile	62	160	59	1.1	2.9	2
China	766	1,646	598	1.5	3.3	2
China, Hong Kong	980	3,595	810	1.6	5.8	4
Colombia	22	50	21	1.0	2.4	0
Croatia	46	41	36	1.5	1.4	3
Czech Republic	69	68	64	1.2	1.2	3
Denmark	4,489	16,433	3,062	2.3	8.2	2,3
Finland	7,003	27,521	4,545	2.7	10.7	3,1
France	51,491	212,668	32,716	7.2	29.9	0
Germany	147,786	599,389	77,121	17.4	70.4	3
Greece	84	150	80	1.1	1.9	1
Hungary	1,516	5,192	1,108	2.0	6.8	1
India	693	711	460	2.1	2.1	3,2
Indonesia	27	55	25	1.1	2.3	3,2
Ireland	639	2,908	581	1.3	5.8	2,1
Israel	5,133	22,119	4,152	2.4	10.3	4,3
Italy	21,746	78,787	15,574	4.0	14.5	2
Japan	440,855	2,373,939	155,127	50.4	271.4	3,1
Malaysia	95	245	80	1.3	3.2	3
Mexico	404	1,032	337	1.3	3.4	0
Netherlands	9,328	34,495	7,240	2.5	9.3	3
New Zealand	780	2,551	710	1.3	4.1	4
Norway	2,174	7,613	1,820	1.6	5.8	2
Philippines	18	39	18	1.0	2.2	1
Poland	286	860	267	1.2	3.5	1
Portugal	60	207	56	1.1	3.8	1
Romania	55	179	53	1.1	3.6	1,2
Russian Federation	547	814	474	1.5	2.2	3,2,1
Saudi Arabia	35	46	33	1.1	1.4	3
Singapore	1,006	3,604	558	2.3	8.2	3
Slovenia	63	92	60	1.1	1.6	3
South Africa	1,256	5,254	1,145	1.4	5.7	3
South Korea	23,221	60,968	7,240	8.7	22.8	3
Spain	2,011	5,141	1,755	1.5	3.9	2
Sweden	16,761	71,194	11,463	3.5	14.8	2,1
Switzerland	18,620	78,961	13,155	3.7	15.5	1
Taiwan	16,726	49,202	7,571	6.6	19.4	2
Thailand	52	113	48	1.1	2.4	3,2
Turkey	32	124	30	1.1	4.1	2
Ukraine	70	88	63	1.1	1.4	3,2
United Kingdom	43,056	211,786	30,315	6.2	30.5	4,3
Venezuela	322	1,028	218	1.6	5.1	3
Others (includes countries with less than 15 patents)	159	404	146	1.1	2.6	

difference between these two differences, therefore, captures the causal effect of the creditor rights change on innovation.

The difference-in-difference test has a number of attractive features. First, it is not subject to the criticism that country- or industry-level unobserved factors influencing innovation are correlated with the level of creditor rights in a country.

Table 2
Countries that underwent a change in creditor rights

Country code	Country name	Year of change
Panel A: Countries that underwent decreases in creditor rights		
ATX	Austria	1982
CAX	Canada	1992
FIX	Finland ¹	1993
IDX	Indonesia	1998
IEX	Ireland	1990
ILX	Israel	1995
INX	India	1993
SEX	Sweden	1995
RUX	Russian Federation	1998
Panel B: Countries that underwent increases in creditor rights		
DKX	Denmark	1984
GBX	United Kingdom	1985
LTX	Lithuania	1995, 1998
ROX	Romania	1999
RUX	Russian Federation	1994

This table shows the list of countries that underwent a change in their creditor rights during the period 1978–1999. From the list of countries undergoing creditor rights changes, as documented in Appendix B of Djankov, McLeish, and Shleifer (2007), we exclude those changes that occurred after the year 1998. This is to allow for at least a few years of patent data after the creditor rights change was effected. From the list of changes in Djankov, McLeish, and Shleifer (2007), we exclude the changes in Armenia, Kazakhstan, Mongolia, and Niger. For these countries, either the country does not have an entry in the patent data or the creditor rights index does not exist for a large part of the sample period.

This is because these tests exploit within-country differences before and after the creditor rights change vis-à-vis similar before–after differences in countries that never experienced such a change. Second, the time-series tests provide point estimates of the effect of changes in creditor rights on innovation using experiments of greatest relevance to policies concerned with promoting innovation.

1.3.1 Basic test. We implement the difference-in-difference test using the following regression:

$$y_{ict} = \beta_i + \beta_c + \beta_t + \beta_1 \delta_{ct} + \beta_2 \cdot Innovation\ Intensity_{i,t-1} + \beta X + \varepsilon_{ict}, \quad (1)$$

where y is the natural logarithm of a measure of innovation for the USPTO patent class (i), country (c), and the year when the patent was applied for (t). For a country c that underwent a creditor rights increase in year m , δ_{ct} equals zero (one) for the years before (after) the change, i.e., for $t \leq m(t \geq m + 1)$. In contrast, for a country c that underwent a creditor rights decrease in year m , δ_{ct} equals one (zero) for the years before (after) the change, i.e., for $t \leq m(t \geq m + 1)$.¹² For countries that did not experience a creditor rights change,

¹² To account for the two-point creditor rights decrease in Finland, we code δ_{ct} to be two before the change and zero after the change for Finland. Since Russia underwent a creditor rights increase in 1994 and a decrease in 1998, for Russia we code δ_{ct} to be equal to zero till 1994, one for the years 1995–1998, and zero thereafter. For Lithuania, which underwent a creditor rights increase in 1995 and in 1998, we code δ_{ct} to be equal to two until 1994, one for the years 1995–1998, and zero thereafter.

δ_{ct} always equals zero. $\beta_i, \beta_c, \beta_t$ denote patent class, country, and application year fixed effects, while X denotes the set of control variables. Since δ_{ct} is defined to be one (zero), one year after the change for countries that decreased (increased) creditor rights, β_1 measures the difference-in-difference effect a year after the change. The innovation intensity for patent class i in year $(t - 1)$, $InnovationIntensity_{i,t-1}$, is measured as the median number of patents applied by U.S. firms in patent class i in year $(t - 1)$. As justified before, instead of using a fixed time window to classify industries based on their propensity to patent, we use a moving window to measure the innovation intensity so as to capture the intertemporal changes in the propensity to innovate caused by technological shocks. Since our proxy for innovation intensity is defined using patents issued to U.S. firms, we exclude all patents issued to U.S. firms (assignee code equal to 3) in y_{ict} . In these and all our other regressions, the standard errors are robust to heteroskedasticity and autocorrelation and are clustered by country. Given our hypothesis that stronger creditor rights lead to lower innovation, we predict that $\beta_1 < 0$.

Notice that compared to the usual difference-in-difference specification, which contains dummies for treatment groups and treatment periods only, including dummies for all the countries, all the application years, as well as the patent classes leads to a much stronger test since we are able to control for time-invariant country and patent class-specific determinants of innovation, as well as the time-varying effects that are common to all countries and patent classes. The application year fixed effects also enable us to control for the problem stemming from the truncation of citations, i.e., citations to patents applied for in later years would, on average, be lower than citations to patents applied for in earlier years. Similarly, the patent class fixed effects also enable us to control for time-invariant differences in patenting and citation practices across industries.

Columns 1–3 in Panel A of Table 3 present the results of this difference-in-difference test without any control variables. In all our tests, we employ in consecutive columns the logarithm of number of patents, the number of citations, and the number of patenting firms as the dependent variable. We find that across Columns 1–3, β_1 is strongly negative.

1.3.2 Controlling for other determinants. Since we use U.S. patents to proxy international innovation, this may introduce biases. Define the difference between the “true” level of innovation in a country and innovation as measured by USPTO patents and citations as the “USPTO bias.” Given the country, patent class, and application year fixed effects in Equation (1), the difference-in-difference coefficient β_1 would be biased only if time-varying omitted variables at the country/ patent class level that affects the USPTO bias are also correlated with the changes in creditor rights. Nevertheless, we control for potential determinants of the USPTO bias. Columns 4–6 of Panel A of Table 3 present results of these tests.

Table 3
Impact of changes in creditor rights on innovation

Dependent variable is logarithm of	(1) Number of patents	(2) Number of citations	(3) Number of firms	(4) Number of patents	(5) Number of citations	(6) Number of firms	(7) Number of patents	(8) Number of citations	(9) Number of firms
Panel A: Main tests									
Creditor rights change	−0.087** (2.17)	−0.121** (2.39)	−0.077** (2.23)	−0.102** (2.25)	−0.143** (2.46)	−0.088** (2.30)			
Creditor rights change (≥2)							−0.099** (2.12)	−0.138** (2.38)	−0.089** (2.25)
Creditor rights change (0,1)							−0.020 (1.08)	−0.053 (1.59)	−0.013 (0.73)
Creditor rights change (−2,−1)							0.041** (2.33)	0.052 (1.53)	0.042** (2.30)
Innovation intensity	0.042*** (3.96)	−0.022 (1.04)	0.023** (2.18)	0.043*** (4.07)	−0.022 (1.01)	0.023** (2.26)	0.043*** (4.11)	−0.021 (0.99)	0.024** (2.28)
Log of GDP per capita				0.665 (1.26)	0.962 (1.60)	0.462 (1.15)	0.649 (1.23)	0.939 (1.57)	0.448 (1.12)
Value added in 3-digit ISIC in a year/total value added by the country in that year				−0.000 (0.95)	−0.004* (1.70)	−0.000 (1.22)	−0.000 (0.96)	−0.006* (1.70)	−0.000 (1.23)
Log of country's imports to the U.S. in the 3-digit ISIC industry in a year				−0.001 (1.48)	0.001 (0.79)	−0.000 (0.49)	−0.001 (1.47)	0.001 (0.80)	−0.000 (0.48)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year				−0.001 (1.36)	−0.001 (1.17)	−0.000 (0.59)	−0.001 (1.37)	−0.001 (1.18)	−0.000 (0.60)
U.S. patent class dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Application year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	86,308	86,308	86,308	86,308	86,308	86,308	86,308	86,308	86,308
Adjusted R-squared	0.60	0.61	0.63	0.60	0.61	0.63	0.60	0.61	0.63

Table 3
(Continued)

Panel B: Excluding countries that underwent a change in patent protection during our sample period (Austria, Indonesia, Ireland, Israel, and Russia)									
Creditor rights change	-0.095** (2.12)	-0.117** (2.20)	-0.077** (2.01)	-0.121** (2.23)	-0.155** (2.34)	-0.095** (2.12)			
Creditor rights change (≥ 2)							-0.118** (2.13)	-0.157** (2.43)	-0.093** (2.12)
Creditor rights change (0,1)							-0.015 (0.48)	0.009 (0.21)	-0.013 (0.56)
Creditor rights change (-2,-1)							-0.025 (0.81)	0.102 (1.18)	-0.023 (0.86)
Innovation intensity	0.043*** (3.89)	-0.018 (0.82)	0.021* (1.98)	0.044*** (4.01)	-0.017 (0.78)	0.022** (2.07)	0.044*** (4.04)	-0.016 (0.76)	0.022** (2.09)
Log of GDP per capita				0.759 (1.33)	1.062 (1.62)	0.511 (1.16)	0.740 (1.30)	1.041 (1.60)	0.496 (1.14)
Value added in 3-digit ISIC in a year/total value added by the country in that year				-0.000 (0.13)	-0.000 (0.88)	-0.000 (0.38)	-0.000 (0.13)	-0.000 (0.88)	-0.000 (0.38)
Log of country's imports to the U.S. in the 3-digit ISIC industry in a year				-0.001 (1.52)	0.000 (0.65)	-0.000 (0.74)	-0.001 (1.51)	0.000 (0.68)	-0.000 (0.73)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year				-0.001 (1.58)	-0.001 (1.29)	-0.000 (0.92)	-0.001 (1.58)	-0.001 (1.28)	-0.000 (0.93)
U.S. patent class dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Application year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	80,440	80,440	80,440	80,440	80,440	80,440	80,440	80,440	80,440
Adjusted R-squared	0.60	0.62	0.63	0.60	0.62	0.63	0.60	0.62	0.63

The OLS regressions above implement the following difference-in-difference test:

$$y_{ict} = \beta_i + \beta_c + \beta_t + \beta_1 \delta_{ct} + \beta_2 \text{InnovationIntensity}_{i,t-1} + \beta X + \varepsilon_{ict},$$

where i , c , t refer to USPTO patent class i , country c , and application year t , respectively. The sample includes patents issued by the USPTO to non-U.S. firms over the period 1978–2002. Innovation intensity for USPTO patent class i is measured as the median number of patents held by a U.S. firm in patent class i in the year $t - 1$. δ_{ct} equals 1 for country c and years $t \geq m + 1$ (years $t < m + 1$) if a creditor rights reform initiated in year m increased (decreased) the rights provided to creditors; δ_{ct} equals 0 otherwise. The sample of changes in creditor rights in different countries is as displayed in Table 4. β_i , β_c , β_t denote patent class, country, and application year dummies. *Creditor rights change* ($-2, -1$) equals δ_{ct} for the years $m - 2$ to $m - 1$, and is zero otherwise. Similarly, *creditor rights change* (0,1) equals δ_{ct} for the years m to $m + 1$ and zero otherwise, while *creditor rights change* (≥ 2) equals δ_{ct} for the years $m + 2$ and beyond, and zero otherwise. The numbers in parentheses denote the t -statistics computed using robust standard errors that are clustered by country. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

An important determinant of the USPTO bias is the extent of bilateral trade that a country has with the United States. Countries that export to the United States would file more patents with the USPTO, particularly in the more innovation-intensive industries. MacGarvie (2006) finds that citations to a country's patent are correlated with the level of exports and imports that the country has with the United States. Therefore, in Equation (1), we add for each country the logarithm of the level of imports and the level of exports that the country has with the United States in each year at each three-digit ISIC industry level, using data from Nicita and Olarreaga (2006).¹³ Another important determinant of patenting by firms in a country may be the comparative advantage that the country possesses in its different industries. We employ, as our proxy for industry-level comparative advantage, the ratio of value added in a three-digit ISIC industry level in a particular year to the total value added by that country in that year. The data for these measures come from United Nations Industrial Development Organization (UNIDO)'s statistics. Finally, since richer countries may innovate more and may also file more patents with the United States, we also include the logarithm of real GDP per capita.

We find in Columns 4–6 of Panel A of Table 3 that none of the control variables are consistently significant in these difference-in-difference tests. Crucially, however, β_1 continues to be negative and statistically significant. We find β_1 to be highest when the dependent variable is the logarithm of citations and lowest when it is the number of firms.

1.3.3 Causality or reverse causality? It is important to understand what caused the changes in creditor rights. Was it the case that creditor rights changed for reasons other than promoting growth and innovation, so that our evidence above can be interpreted truly as a causal effect of the change on innovation? Or, was it the case that creditor rights changes were part of an overall package to promote or give an extra boost to growth and innovation, so that the evidence above exhibits some reverse causality? Note that in either of these cases, the evidence lends support to our claim that creditor rights can affect the extent of innovative activity. Nevertheless, we examine reverse causality in Columns 7–9 of Panel A of Table 3 by examining the dynamic effect of these creditor rights changes. In Section 1.3.5, we describe the causes of the creditor rights changes in each country.

If the creditor rights change was effected to provide an extra boost to growth and innovation already occurring due to some other changes in the economy, then we might see an “effect” of the change even prior to the change itself. To examine this, we follow Bertrand and Mulainathan (2003) in decomposing our *creditor rights change* variable into three separate time periods: (i)

¹³ We match the patent classes to the three-digit ISIC using a two-step procedure: first, the updated NBER patent data set (patsic02.dta on Brownwyn Hall's homepage) assigns each patent to a two-digit SIC. We then employed the concordance from two-digit SIC to three-digit ISIC codes. Since every patent is already assigned to a patent class in the original NBER patent data set, this completes our match from the patent class to the three-digit ISIC code.

creditor rights change $(-2, -1)$, which captures any effects from two years before to a year before the change; (ii) *creditor rights change* $(0, 1)$, which captures the effect in the year of the change and the year after the change; and (iii) *creditor rights change* (≥ 2) , which captures the effect two years after the change and beyond. If the coefficient of *creditor rights change* $(-2, -1)$ is negative and statistically significant, that may be symptomatic of reverse causation. However, we find that the coefficients are positive and statistically significant in two of the three specifications for *creditor rights change* $(-2, -1)$, negative but statistically insignificant for *creditor rights change* $(0, 1)$, and negative and statistically significant for *creditor rights change* (≥ 2) . In fact, the coefficient estimates for *creditor rights change* (≥ 2) are consistently higher than the estimate of β_1 in Columns 1–6 of Table 3. The coefficient of *creditor rights change* $(-2, -1)$ being positive and significant in two of the three specifications suggests that innovation increased two years before the creditor rights change, which is puzzling. We investigate this effect in greater detail in Section 1.3.4. The significant negative effect of creditor rights changes being observed only two years after the change is consistent with evidence in Kondo (1999) that there is a one-and-a-half year lag between patent applications and R&D investment.

1.3.4 Country-level changes in patent protection. Country-level changes in the protection provided to patents affect the ex ante incentives of firms to pursue innovative projects and, in turn, their filing of patents with the USPTO. Overlapping with the time of the creditor rights changes in our sample, Austria, Indonesia, Ireland, and Russia increased the length of patent protections in their countries from 18 to 20 years, from 14 to 20 years, from 16 to 20 years, and from 15 to 20 years, respectively, while Israel introduced special patent protection for pharmaceuticals. In fact, in Indonesia, Ireland, and Russia, the increase in patent protection preceded by a few years the creditor rights change.¹⁴ To ensure that the difference-in-difference results above are not being driven by changes in patent protection, we reestimate them by excluding these countries from our sample. Panel B of Table 3 replicates the results of Panel A using this reduced sample. We note that the estimate of β_1 is strongly negative in Columns 1–6 of Panel B. Interestingly, we find by comparing Columns 7–9 of Panel B to those in Panel A that the coefficient of *creditor rights change* $(-2, -1)$ is not statistically significant once we exclude the countries that experienced increases in patent protection. In Columns 7–9 of Panel B, we find that only the coefficient of *creditor rights change* (≥ 2) is negative and significant.

1.3.5 Causes of creditor rights changes. We now discuss the reasons behind the changes in the creditor rights in our “treatment” sample of countries.

¹⁴ We thank Francesca Cornelli for sharing these data with us.

The weakening of creditor rights in Israel in 1995 was precisely to provide entrepreneurship a boost. It represented a greater tolerance toward debt undertaking and the overextended borrowers who fell into financial trouble. In fact, “the changing orientation of Israel’s economy from being socialist-based to more capitalistic can also be linked to the liberalization of the fresh-start policy. As entrepreneurship became a more widely accepted activity in Israel, society began to acknowledge the incentives a more liberal fresh-start policy could provide to a private market economy” (Erfat 2000).

Other changes have not necessarily occurred to promote such fresh starts. For instance, the primary objective of the weakening of creditor rights in Canada in 1992 was to increase the chances of survival of businesses that were experiencing financial difficulties, and, as a consequence, to save jobs (Martel 1994), an *ex post* objective rather than an *ex ante* one to promote innovation *per se*. In the case of India, the motivation for weakening of creditor rights in 1993 was to protect the domestic, uncompetitive firms that had been forced into bankruptcy by the deregulation and introduction of foreign competition in 1991 (Kang and Nayar 2004), a lobbying-based outcome rather than one aimed at efficiency.

In contrast, the weakening of creditor rights in Finland in 1993 (by two points) and Indonesia in 1998 were prompted by the severity of ongoing crises. In the case of Finland, the real GDP had dropped by about 14% and unemployment had risen from 3% to nearly 20%. The creditor rights were, however, part of a larger stimulus package in both countries. In Finland, restrictions on foreign ownership were completely abolished and the accounting legislation was improved. These measures were attributed to the acceleration in development of the stock market and the venture capital activity, and the rebounding of employment rates (Hyytinen, Kuosa, and Takalo 2001). In Indonesia, bankruptcy law reforms included secured transactions law reforms and reforms to anticorruption legislations.

Finally, the weakening of creditor rights in the United Kingdom in 1985 was largely to mirror the success of the U.S. Chapter 11 bankruptcy in providing a formal structure for reorganization of solvent but illiquid institutions (Armour, Cheffins, and Skeel 2002), whereas their strengthening in Russia in 1994, Lithuania in 1995, and Romania in 1999 was a part of their transition and was viewed as a way to boost lending (Haselmann, Pistor, and Vig 2009) and make more efficient the bankruptcy systems ridden by inexperienced judges.

These examples illustrate the following important points. First, creditor right changes have sometimes been introduced precisely to promote growth and innovation. Second, these changes have often arisen due to lobbying and job-saving objectives, exogenous to the issue of promoting innovation. And, third, these changes have also often been timed to turn around economies that are in crises or at the verge of growth spurts, but importantly, creditor right changes have been an important part of the overall stimulus package in these cases. These facts, along with our empirical tests on reverse causality, give us confidence

that the relationship between creditor right changes and innovation unearthed in our time-series tests is indeed economically meaningful and causal.

1.3.6 Economic magnitudes. The economic magnitude of the effect of creditor rights change on innovation is quite significant. For this purpose, we use Columns 1–3 of Panel A of Table 3. Compared to the counties that did not undergo a creditor rights change, the group of countries that underwent a creditor rights decrease generated 10.7% more patents, 15.4% more citations to these patents, and 9.2% more patenting firms. In contrast, the group of countries that underwent a creditor rights increase generated 9.7% less patents, 13.3% less citations to these patents, and 8.4% less patenting firms than those countries that did not experience a creditor rights change.

1.3.7 The effect of creditor rights changes on innovation by a typical firm. In Table 4, we display the results of difference-in-difference tests for innovation done by a typical firm in our sample. We employ as our measures of innovation the median (Columns 1 and 4), the average (Columns 2 and 5), and the citation-weighted average (Columns 3 and 6) number of patents held by a firm in a country, patent class, and year. We find in Table 4 that β_1 is strongly negative and statistically significant. The economic magnitude for these average measures of innovation is lower than for the aggregate measures of innovation in Table 3.

Taken together, Tables 3 and 4 suggest that the decrease (increase) in creditors rights in a country caused more (less) innovation at both the extensive and intensive margins.

1.4 The causal mechanism

Having examined the direct effect of creditor rights on innovation, we now describe our tests of the causal mechanism that leads to the negative effect of creditor rights on innovation. We test the hypothesis that the difference in innovation between more innovation-intensive industries (such as biotechnology) and less innovation-intensive industries (such as apparel and textiles) would decrease with the strength of creditor rights. We first examine this hypothesis using cross-sectional tests and then using the changes in creditor rights.

1.4.1 Cross-sectional tests. The regression below estimates inter-industry differences in innovation across different bankruptcy regimes:

$$y_{ict} = \beta_i + \beta_c + \beta_t + \beta_1 \cdot (CreditorRights_{ct} \cdot InnovationIntensity_{i,t-1}) + \beta_2 \cdot CreditorRights_{ct} + \beta_3 \cdot InnovationIntensity_{i,t-1} + \beta X + \varepsilon_{ict}, \quad (2)$$

where y is the natural logarithm of a measure of innovation for patent class (i), country (c) in year (t). $CreditorRights_{ct}$ is a measure of country c 's strength of creditor rights in year t . As in Equation (1), $InnovationIntensity_{i,t-1}$ equals the

Table 4
Impact of changes in creditor rights on average innovation

Dependent variable is logarithm of	(1) Median patents	(2) Average patents	(3) Citation weighted average patents	(4) Median patents	(5) Average patents	(6) Citation weighted average patents
Creditor rights change	−0.032** (2.14)	−0.046* (1.95)	−0.111*** (2.88)	−0.035** (2.27)	−0.051** (2.10)	−0.120*** (2.99)
Innovation intensity	0.012* (1.82)	0.020*** (2.96)	−0.048*** (3.11)	0.012* (1.83)	0.020*** (2.96)	−0.047*** (3.11)
Log of GDP per capita				0.094* (1.82)	0.212 (1.60)	0.387* (1.69)
Value added in 3-digit ISIC in a year/total value added by the country in that year				0.000 (0.89)	−0.000 (0.06)	−0.000 (0.93)
Log of country's imports to the U.S. in the 3-digit ISIC industry in a year				−0.001*** (3.90)	−0.001*** (2.93)	0.001* (1.75)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year				−0.000 (0.05)	−0.000* (1.94)	−0.000 (0.25)
U.S. patent class dummies	Yes	Yes	Yes	Yes	Yes	Yes
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Application year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	86308	86308	86308	86308	86308	86308
Adjusted <i>R</i> -squared	0.11	0.28	0.48	0.11	0.28	0.49

The OLS regressions above implement the following difference-in-difference test:

$$y_{ict} = \beta_i + \beta_c + \beta_t + \beta_1 \delta_{ct} + \beta_2 \text{InnovationIntensity}_{i,t-1} + \beta X + \varepsilon_{ict},$$

where i , c , t refer to USPTO patent class i , country c , and application year t , respectively. The sample includes patents issued by the USPTO to non-U.S. firms over the period 1978–2002. Innovation intensity for USPTO patent class i is measured as the median number of patents held by a U.S. firm in patent class i in the year $t - 1$. δ_{ct} equals 1 for country c and years $t \geq m + 1$ (years $t < m + 1$) if a creditor rights reform initiated in year m increased (decreased) the rights provided to creditors. δ_{ct} equals 0 otherwise. The sample of changes in creditor rights in different countries is as displayed in Table 4. β_i , β_c , β_t denote patent class, country, and application year dummies. *Creditor rights change* (−2, −1) equals δ_{ct} for the years $m - 2$ to $m - 1$, and is 0 otherwise. Similarly, *creditor rights change* (0, 1) equals δ_{ct} for the years m to $m + 1$ and 0 otherwise, while *creditor rights change* (≥ 2) equals δ_{ct} for the years $m + 2$ and beyond, and 0 otherwise. The numbers in parentheses denote the t -statistics computed using robust standard errors that are clustered by country. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

median number of patents applied by U.S. firms in patent class i in year $(t - 1)$ and y_{ict} excludes all patents issued to U.S. firms.

Since we are primarily interested in the coefficient of the interaction of country-level creditor rights with innovation intensity at the patent class level, we include country-fixed effects (β_c) and patent class fixed effects (β_i) in all our specifications. This enables us to control for all time-invariant unobserved heterogeneities at the country and patent class levels. The application year fixed effects (β_t) control for intertemporal differences, as well as the problem stemming from the truncation of citations.

Our hypothesis is that $\beta_1 < 0$. Columns 1–3 in Panel A of Table 5 show the results of the test of Equation (2). We find that for each of the three dependent variables, β_1 is negative and statistically significant.¹⁵

Controlling for bilateral trade, comparative advantages, and GDP. We now control for the effects of (i) the extent of bilateral trade that a country has with the United States; (ii) comparative advantages that a country has in its different industries; and (iii) the country's GDP per capita. Note that unlike the difference-in-difference tests above, differences in industry-level comparative advantages across different countries are an important concern in the cross-sectional tests since these affect our interpretation of β_1 in Equation (2) as the relative effect of creditor rights on innovation. To see this, consider two countries with different level of creditor rights: call the country with lower creditor rights country 1 and the one with higher creditor rights country 2. Suppose country 1 and the United States have a comparative advantage in biotechnology, while country 2 (which has higher creditor rights) possesses a comparative advantage in tangible, asset-intensive industries such as cement or steel, which have a lower propensity to patent. Now, even if our empirical results showed that $\beta_1 < 0$, this could be consistent with the interpretation that country 1 and the United States innovate in biotechnology, while country 2 innovates in cement or steel. However, since firms in the cement and steel industries have a lower propensity to patent, we obtain that $\beta_1 < 0$. Such an interpretation would imply that there is no relative effect of creditor rights on innovation, other than a spurious one.

The measures for bilateral trade and comparative advantage are as described in Section 1.3.2. Since these variables may be correlated with the interaction of creditor rights and innovation intensity, we interact each of them with $InnovationIntensity_{i,t-1}$ as well.

¹⁵ We also check this result separately for the periods 1978–1990 and 1991–2002. Furthermore, we also test by excluding those countries for which the aggregate number of patents over the period 1978–2002 is less than 100. Our results remain as strong for these subsamples. We also perform two different robustness checks for the definition of innovation intensity. First, we verify that the results are unchanged if we compute innovation intensity as the median number of patents applied by U.S. firms in the year $(t - 2)$. We use the year $(t - 2)$ since we find using the de-trended number of patents and citations for U.S. firms that both follow an AR(1) process. Second, we test by constructing innovation intensity using U.S. patents filed by Japanese firms and excluding Japanese patents from the analysis. Our results are unaltered in both cases.

Table 5
Relative impact of creditor rights on aggregate innovation in different industries

Dependent variable is logarithm of	(1) Number of patents	(2) Number of citations	(3) Number of firms	(4) Number of patents	(5) Number of citations	(6) Number of firms
Panel A: Main tests						
Creditor rights × innovation intensity	−0.040** (2.36)	−0.039** (2.14)	−0.027* (1.88)	−0.038** (2.42)	−0.030* (1.95)	−0.028** (1.99)
Creditor rights	0.011 (0.16)	0.134 (0.79)	−0.002 (0.04)	−0.013 (0.18)	0.096 (0.57)	−0.016 (0.29)
Innovation intensity	0.139*** (2.70)	0.074 (1.29)	0.089* (1.88)	0.535 (1.01)	0.186 (0.21)	1.035** (2.41)
Log of GDP per capita				0.671 (1.15)	0.872 (1.44)	0.533 (1.21)
Value added in 3-digit ISIC in a year/total value added by the country in that year				0.008 (0.74)	−0.010 (0.75)	0.004 (0.52)
Log of country's imports to the U.S. in the 3-digit ISIC industry in a year				0.000 (0.23)	0.003 (1.15)	0.001 (1.20)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year				0.004*** (3.03)	0.005** (2.17)	0.004** (2.45)
Log of GDP per capita × innovation intensity				−0.036 (0.69)	−0.007 (0.07)	−0.091** (2.24)
Value added in 3-digit ISIC in a year/total value added by the country in that year × innovation intensity				−0.010 (0.99)	0.005 (0.36)	−0.007 (0.79)
Log of country's imports to the U.S. in the 3-digit ISIC industry in a year × innovation intensity				−0.001 (0.62)	−0.002 (1.01)	−0.001 (1.16)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year × innovation intensity				0.004*** (3.22)	0.005** (2.35)	0.004** (2.51)
U.S. patent class dummies	Yes	Yes	Yes	Yes	Yes	Yes
Country dummies	Yes	Yes	No	Yes	Yes	No
Application year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	85,300	85,300	83,255	85,300	85,300	83,255
Adjusted <i>R</i> -squared	0.60	0.60	0.61	0.61	0.62	0.63

Table 5
(Continued)

Panel B: Tests controlling for legal origin and other time-varying unobserved factors at country and patent category levels

Creditor rights × innovation intensity	−0.011*	−1.107**	−0.018*	−0.047***	−0.141***	−0.031***	−0.045***	−0.138***	−0.031***
	(1.66)	(2.22)	(1.72)	(3.74)	(2.69)	(3.24)	(3.58)	(2.62)	(3.18)
Creditor rights	0.001	−0.757	−0.017				0.069***	−0.107	0.057***
	(0.15)	(1.27)	(1.32)				(3.87)	(1.43)	(4.13)
Innovation intensity	0.154***	−1.741	0.033	0.162***	0.286**	0.104***	0.145***	0.232	0.098***
	(4.86)	(0.71)	(0.64)	(4.71)	(1.97)	(3.89)	(4.17)	(1.58)	(3.60)
One if English legal origin × innovation intensity	−0.007	3.498	0.004						
	(0.22)	(1.42)	(0.08)						
One if French legal origin × innovation intensity	0.014	4.234*	0.095*						
	(0.47)	(1.79)	(1.87)						
One if German legal origin × innovation intensity	−0.027	4.498*	−0.004						
	(0.89)	(1.95)	(0.09)						
U.S. patent class dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Application year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
(Country × application year) dummies	No	No	No	Yes	Yes	Yes	No	No	No
(Patent category × application year) dummies	No	No	No	No	No	No	Yes	Yes	Yes
Observations	83,255	83,255	83,255	68,457	68,457	68,457	68,442	68,442	68,442
Adjusted R-squared	0.29	0.41	0.58	0.59	0.54	0.62	0.58	0.52	0.61

The OLS regressions above implement the following model:

$$y_{ict} = \beta_i + \beta_c + \beta_t + \beta_1(\text{CreditorRights}_{ct} * \text{InnovationIntensity}_{i,t-1}) + \beta_2 \text{CreditorRights}_{ct} + \beta_3 \text{InnovationIntensity}_{i,t-1} + \varepsilon_{ict},$$

where i, c, t refer to USPTO patent class i , country c , and application year t , respectively. The sample includes patents issued by the USPTO to non-U.S. firms over the period 1978–2002. Innovation intensity for USPTO patent class i is measured as the median number of patents held by a U.S. firm in patent class i in the year $t - 1$. The creditor rights index for country c is from Djankov, McLiesh, and Shleifer (2007)—a higher measure indicates stronger rights provided to creditors in bankruptcy. $\beta_i, \beta_c, \beta_t$ denote patent class, country, and application year dummies. The numbers in parentheses denote the t -statistics computed using robust standard errors that are clustered by country. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Columns 4–6 of Panel A of Table 5 present the results of adding these controls to Equation (2). We find that among these controls, only exports display a significant effect—countries that export more, file more patents with the United States, and particularly so in the innovation-intensive industries. Importantly, β_1 continues to be negative and statistically significant. The coefficient decreases when the dependent variable is the logarithm of the number of patents or number of citations, but increases when it is the number of firms.

Controlling for other unobserved heterogeneities. As additional tests for the interaction effect in Equation (2), we control for other forms of unobserved heterogeneities. First, we include interactions of our proxy for innovation intensity with the legal origin of the country to capture omitted variables at the country–patent class levels. We also interacted other country-level variables documented in the law and finance literature (La Porta et al. 1998), such as the rule of law, enforcement of contracts, an index of public legal enforcement, and efficiency of the bankruptcy procedure with our proxy for innovation intensity. Since only the legal origin interactions were statistically significant, we report them in Columns 1–3 of Panel B of Table 5. Second, in Columns 4–6 and Columns 7–9 of Panel B, we control for time-varying unobserved factors at the country and patent class levels, respectively, by including (country $\times$ time) and (industry $\times$ time) fixed effects. Across all these columns, we find that β_1 stays negative and statistically significant.

Controlling for the effect of financial development. Despite our inclusion of a country's GDP per capita, other country-level variables, country and (country $\times$ time) fixed effects, it is possible that the weakness of creditor rights may actually be capturing the effect of financial development, which has been argued to boost innovation through greater competition (Romer 1986; Grossman and Helpman 1991; Aghion and Howitt 1992). For example, creditor rights may be designed to be stronger to boost credit and intermediation in poorly developed financial markets. Another orthogonal concern is that if financial contracts allow firms to get around the inefficiencies imposed by the bankruptcy code, then the effect of the code on real choices of firms would be muted. Since richness of contracting is likely related to the level of financial development (measured, for example, by quality of accounting standards), it may be important to control for variations in this level across countries.

To address these issues, we employ two measures of financial development: accounting standards and logarithm of private credit to GDP per capita (from La Porta et al. 1998; Rajan and Zingales 1998).¹⁶ Requiring these variables

¹⁶ Our results here are robust to the use of two other measures of financial development: (i) total capitalization to GDP is the ratio of the sum of equity market capitalization (as reported by the IFC) and domestic credit (IFS lines 32a–32f but not 32e) to GDP from Rajan and Zingales (1998), and (ii) domestic private credit to GDP is the ratio of domestic credit to the private sector, which is from IFS line 32d, over GDP from Rajan and Zingales (1998).

reduces our sample size to around 40 countries. In Table 6, we show the results of tests including financial development and its interaction with innovation intensity.¹⁷ We find that β_1 continues to be strongly negative.

Economic magnitude of the effect. The economic magnitude of the effect of the interaction term is quite significant. We use Columns 1–3 in Panel A of Table 5 to estimate these economic magnitudes. Consider two patent classes that differ by one patent per year in the median number of patents filed by U.S. firms in that patent class. If the creditor rights index is zero, then the more innovative patent class would generate 14.9% more patents in a year than the less innovative one. If, instead, the creditor rights index is one, then this difference is 10.4%. When the creditor rights index equals its maximum value of four, the more innovative-intensive patent class generates 2.1% less patents in a year than the less innovative one. So, when we go from a country with a creditor rights index of 0 to 1, the difference in innovation between two adjacent patent classes reduces by more than 30% and reverses when the index equals 4. The economic magnitudes for the number of citations and the number of firms are of a similar order of magnitude.

The effect of creditor rights on innovation by a typical firm. In Table 7, we display the results of the cross-sectional tests for the innovation done by the typical firm in our sample. As in Table 4, we employ as our measure of innovation the median-, average-, and citation-weighted average number of patents held by a firm in a country, patent class, and year. We find that β_1 is strongly negative, though the economic magnitudes for these average measures are lower than that for the aggregate measures in Table 5. Taken together, Tables 5 and 7 suggest that the effect of creditor rights on interindustry differences in innovation is present at both extensive and intensive margins.

Discussion of the cross-sectional test. We observe in our estimates of the economic magnitude above that when the creditor rights index equals 4, the less innovation-intensive industry innovates more than the more innovation-intensive industry. This is consistent with the comparative advantage critique that we outlined above, i.e., countries where the creditor rights index equals 4 focus their innovation resources in the less innovation-intensive industries. Thus, despite our attempt to control for the effects of industry-level comparative advantages in different countries, in these cross-sectional tests, our inference of the effect of the interaction of creditor rights with innovation intensity is affected by differences in comparative advantage.

To alleviate concerns about the effects of comparative advantage, we reexamine the interaction effect using exogenous changes in creditor rights.

¹⁷ Since our regressions include country-fixed effects, we estimate the effect of legal origin variables here by aggregating over the country-fixed effects.

Table 6
Relative impact of creditor rights on innovation in different industries after controlling for the effect of financial development

Dependent variable is log of	(1) Number of patents	(2) Number of patents	(3) Number of citations	(4) Number of citations	(5) Number of firms	(6) Number of firms
Which financial development measure?	Accounting standards	Log private credit to GDP per capita	Accounting standards	Log private credit to GDP per capita	Accounting standards	Log private credit to GDP per capita
Creditor rights × median number of patents in U.S. patent class	−0.038*** (3.14)	−0.039*** (3.20)	−0.130** (2.54)	−0.132** (2.53)	−0.025*** (2.70)	−0.023** (2.39)
Creditor rights	0.006 (0.40)	0.008 (0.50)	0.049 (0.78)	0.052 (0.80)	−0.005 (0.47)	−0.008 (0.69)
Median number of patents in U.S. patent class	0.463*** (3.97)	0.139*** (4.22)	1.004** (2.02)	0.266* (1.89)	0.344*** (3.77)	0.086*** (3.32)
Financial development measure × median number of patents in patent class	−0.005*** (2.89)	−0.005 (0.12)	−0.011 (1.54)	−0.032 (0.19)	−0.004*** (2.90)	−0.066** (2.08)
Financial development measure	0.002 (0.35)	0.037 (0.47)	0.080*** (2.93)	0.367 (1.10)	0.002 (0.40)	0.095 (1.55)
One if English legal origin	−0.340*** (3.68)	−0.450*** (3.84)	−0.362 (0.92)	−0.956* (1.91)	−0.212*** (2.93)	−0.340*** (3.70)
One if French legal origin	−0.519*** (2.68)	−0.387*** (4.53)	0.426 (0.51)	−1.051*** (2.89)	−0.370** (2.43)	−0.274*** (4.10)
One if German legal origin	0.668*** (10.85)	0.898*** (30.69)	2.806*** (10.68)	2.019*** (16.18)	0.589*** (12.21)	0.586*** (25.60)
Patent class dummies	Yes	Yes	Yes	Yes	Yes	Yes
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Application year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	83,458	83,458	83,458	83,458	83,458	83,458
Adjusted <i>R</i> -squared	0.55	0.55	0.51	0.51	0.58	0.58

The OLS regressions above add a measure of financial development and the interaction of the measure of financial development with patenting intensity to the basic model examined in Table 2. We use the following proxies for financial development: (1) Accounting standards is an index created by Center for International Financial Analysis and Research examining and rating companies' 1990 annual reports on their inclusion or omission of 90 items from La Porta et al. (1998), (2) Log private credit to GDP per capita is the logarithm of the ratio of domestic private credit (IFS line 32d) to the GDP per capita from La Porta et al. (1998)¹⁴. The dependent variable in the regressions is the logarithm of the total number of patents, the total number of citations to these patents, or the total number of patenting firms. The sample includes patents issued by the USPTO to non-U.S. firms over the period 1978–2002. Patent intensity for a USPTO patent class is measured as the median number of patents held by a U.S. firm in that patent class in the previous year. The creditor rights index for a country is from Djankov, McLiesh, and Shleifer (2007)—a higher measure indicates stronger rights provided to creditors in bankruptcy. The numbers in parentheses denote the *t*-statistics computed using robust standard errors that are clustered by country. *** and ** denote significance at the 1% and 5% levels, respectively.

Table 7
Relative impact of creditor rights on average innovation in different industries

Dependent variable is logarithm of	(1) Median patents	(2) Median patents	(3) Average patents	(4) Average patents	(5) Citation weighted average patents	(6) Citation weighted average patents
Creditor rights × innovation intensity	−0.018*** (3.33)	−0.013** (2.56)	−0.020*** (2.64)	−0.016** (1.99)	−0.311*** (4.27)	−0.329*** (4.46)
Creditor rights	0.013 (1.57)	0.006 (0.62)	0.019 (1.01)	0.009 (0.46)	0.278** (2.44)	0.291*** (2.67)
Innovation intensity	0.130*** (9.66)	−0.429*** (3.40)	0.136*** (7.29)	−0.266* (1.85)	0.137 (1.02)	0.530 (0.22)
Log of GDP per capita		0.029 (0.49)		0.150 (1.13)		0.226 (0.61)
Value added in 3-digit ISIC in a year/total value added by the country in that year		−0.000 (1.14)		−0.000 (0.15)		−0.000 (0.22)
Log of country's imports to the U.S. in the 3-digit ISIC industry in a year		0.000 (0.31)		0.000 (0.02)		−0.019** (2.17)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year		−0.001 (0.54)		−0.000 (0.14)		0.008* (1.69)
Log of GDP per capita × innovation intensity		0.056*** (4.44)		0.040*** (2.80)		−0.048 (0.20)
Value added in 3-digit ISIC in a year/total value added by the country in that year × innovation intensity		0.000 (1.25)		0.000 (0.16)		0.000 (0.21)
Log of country's imports to the U.S. in the 3-digit ISIC industry in a year × innovation intensity		−0.001 (0.64)		−0.000 (0.23)		0.019** (2.21)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year × innovation intensity		0.001 (0.46)		−0.000 (0.12)		0.008* (1.71)
U.S. patent subcategory dummies	Yes	Yes	Yes	Yes	Yes	Yes
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Application year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	85,284	85,284	85,284	85,284	85,284	85,284
Adjusted <i>R</i> -squared	0.06	0.06	0.19	0.19	0.34	0.34

The OLS regressions above implement the following model:

$$y_{ict} = \beta_i + \beta_c + \beta_t + \beta_1(CreditorRights_{ct} \cdot InnovationIntensity_{i,t-1}) + \beta_2 CreditorRights_{ct} + \beta_3 InnovationIntensity_{i,t-1} + \epsilon_{ict},$$

where i , c , t refer to USPTO patent class i , country c , and application year t , respectively. The sample includes patents issued by the USPTO to non-U.S. firms over the period 1978–2002. Innovation intensity for USPTO patent class i is measured as the median number of patents held by a U.S. firm in patent class i in the year $t - 1$. The creditor rights index for country c is from Djankov, McLiesh, and Shleifer (2007)—a higher measure indicates stronger rights provided to creditors in bankruptcy. β_i , β_c , β_t denote patent class, country, and application year dummies. The numbers in parentheses denote the t -statistics computed using robust standard errors that are clustered by country. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

1.4.2 Third-difference test. We now use the creditor rights changes to perform a third-difference test,

$$y_{ict} = [\beta_c + \beta_t + \beta_1 \delta_{ct}] \cdot InnovationIntensity_{i,t-1} + \beta_i + \beta_c + \beta_t + \beta_2 \cdot \delta_{ct} + \beta X + \varepsilon_{ict}, \quad (3)$$

where the variable definitions are identical to those in Equation (1). This specification is equivalent to

$$\frac{\partial y_{ict}}{\partial InnovationIntensity_{i,t-1}} = \beta_c + \beta_t + \beta_1 \delta_{ct}. \quad (4)$$

Thus, β_1 now estimates the relative effect on innovation across different industries as a difference-in-difference. As before, our hypothesis is that $\beta_1 < 0$.

Panel A of Table 8 presents the results of this third-difference test. In Columns 1–3, we include our variables for bilateral trade, industry-level comparative advantages, and the country's GDP per capita. As in the cross-sectional tests, we also interact them with our proxy for innovation intensity. Across Columns 1–3, we find the coefficient β_1 to be strongly negative. As in the cross-sectional tests in Panel A of Table 5, only the coefficient of exports and its interaction with innovation intensity are positive and significant.

As in the direct effect of creditor rights changes, we test for reverse causality by examining the dynamic effect of the change in Columns 4–6 of Table 8. The negative and statistically significant effect on the interaction of innovation intensity with creditor rights change (≥ 2) combined with the insignificant effects for the other two interactions suggests the absence of any reverse causality in the interaction effect too.

In Panel B of Table 8, we reexamine the third difference by excluding countries that underwent a change in patent protection. We find that the coefficient β_1 in Panel B is very similar to that in Panel A, which suggests that the third-difference results are robust to the changes in patent protection in Austria, Indonesia, Ireland, Israel, and Russia.

The economic magnitude of the interaction effect in these time-series tests is quite significant. For this purpose, we use Columns 1–3 of Panel A of Table 8. Since the coefficient of innovation intensity is statistically indistinguishable from zero in Columns 1–6, there is no difference in innovation between two adjacent classes for the control group of countries. For the group of countries that underwent a creditor rights decrease, the more innovative class generated 13.2% more patents, 32.7% more citations to these patents, and 10.8% more patenting firms than its adjacent less innovative class. In contrast, for the group of countries that underwent a creditor rights increase, the more innovative class generated 11.7% less patents, 24.6% less citations to these patents, and 10.8% less patenting firms than its adjacent less innovative class.

Table 8
Relative impact of changes in creditor rights on innovation in different industries

Dependent variable is logarithm of	(1) Number of patents	(2) Number of citations	(3) Number of firms	(4) Number of patents	(5) Number of citations	(6) Number of firms
Panel A: Main tests						
Creditor rights change $\times$ innovation intensity	-0.124** (2.06)	-0.283** (2.20)	-0.103* (1.71)			
Creditor rights change	0.026 (0.28)	0.147 (0.86)	0.019 (0.23)			
Creditor rights change (≥ 2) $\times$ innovation intensity				-0.136* (1.93)	-0.311** (2.37)	-0.118* (1.75)
Creditor rights change (0,1) $\times$ innovation intensity				-0.044 (0.36)	-0.355 (1.38)	0.116 (1.49)
Creditor rights change (-2,-1) $\times$ innovation intensity				0.071 (0.36)	-0.278 (0.92)	0.057 (0.44)
Creditor rights change (≥ 2)				0.041 (0.39)	0.181 (1.04)	0.036 (0.41)
Creditor rights change (0,1)				0.038 (0.31)	0.385 (1.51)	-0.125 (1.50)
Creditor rights change (-2,-1)				-0.096 (0.50)	0.215 (0.70)	-0.080 (0.63)
Innovation intensity	2.020 (0.33)	5.246 (0.53)	3.400 (0.92)	1.928 (0.31)	5.417 (0.55)	3.227 (0.89)
Log of GDP per capita $\times$ innovation intensity	-0.291 (0.43)	-0.689 (0.65)	-0.425 (1.07)	-0.281 (0.42)	-0.705 (0.67)	-0.406 (1.03)
Value added in 3-digit ISIC in a year/total value added by the country in that year $\times$ innovation intensity	-0.000 (0.88)	0.000 (0.37)	-0.000 (0.31)	-0.000 (0.88)	0.000 (0.36)	-0.000 (0.36)
Log of country's imports to the U.S. in the 3-digit ISIC industry in a year $\times$ innovation intensity	-0.001 (0.45)	-0.004 (1.55)	-0.000 (0.41)	-0.001 (0.45)	-0.003 (1.47)	-0.001 (0.50)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year $\times$ innovation intensity	0.004*** (3.00)	0.004* (1.83)	0.004*** (2.89)	0.004*** (3.00)	0.004* (1.96)	0.004*** (2.87)
Log of GDP per capita	0.971 (1.16)	1.679 (1.52)	0.904 (1.37)	0.946 (1.14)	1.673 (1.51)	0.872 (1.34)
Value added in 3-digit ISIC in a year/total value added by the country in that year	0.000 (0.66)	-0.000 (0.73)	0.000 (0.06)	0.000 (0.67)	-0.000 (0.72)	0.000 (0.10)

Table 8
(Continued)

Dependent variable is logarithm of	(1) Number of patents	(2) Number of citations	(3) Number of firms	(4) Number of patents	(5) Number of citations	(6) Number of firms
Log of country's imports to the U.S. in the 3-digit ISIC industry in a year	-0.000 (0.08)	0.004 (1.57)	0.000 (0.23)	-0.000 (0.09)	0.004 (1.50)	0.000 (0.33)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year	0.004*** (2.89)	0.003* (1.68)	0.004*** (2.88)	0.004*** (2.89)	0.004* (1.81)	0.004*** (2.88)
U.S. patent class dummies	Yes	Yes	Yes	Yes	Yes	Yes
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Application year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Country dummies $\times$ innovation intensity	Yes	Yes	Yes	Yes	Yes	Yes
Application year dummies $\times$ innovation intensity	Yes	Yes	Yes	Yes	Yes	Yes
Observations	86,308	86,308	86,308	86,308	86,308	86,308
Adjusted <i>R</i> -squared	0.60	0.61	0.63	0.60	0.61	0.63
Panel B: Excluding countries that underwent a change in patent protection during our sample period (Austria, Indonesia, Ireland, Israel, and Russia)						
Creditor rights change $\times$ innovation intensity	-0.125* (1.72)	-0.237** (2.00)	-0.081* (1.67)			
Creditor rights change	0.008 (0.11)	0.088 (0.73)	-0.011 (0.22)			
Creditor rights change (≥ 2) $\times$ innovation intensity				-0.125* (1.68)	-0.254** (2.15)	-0.091* (1.84)
Creditor rights change (0,1) $\times$ innovation intensity				-0.083 (0.50)	-0.436* (1.67)	0.105 (0.90)
Creditor rights change (-2,-1) $\times$ innovation intensity				0.050 (0.30)	-0.235 (0.94)	0.071 (0.64)
Creditor rights change (≥ 2)				0.010 (0.13)	0.103 (0.85)	0.001 (0.02)
Creditor rights change (0,1)				0.068 (0.40)	0.458* (1.69)	-0.121 (1.00)
Creditor rights change (-2,-1)				-0.079 (0.46)	0.157 (0.60)	-0.098 (0.84)
Innovation intensity	2.020 (0.33)	5.246 (0.53)	3.400 (0.92)	1.928 (0.31)	5.417 (0.55)	3.227 (0.89)

Table 8
(Continued)

Log of GDP per capita × innovation intensity	−0.299 (0.98)	−0.688 (1.39)	−0.472** (2.46)	−0.298 (0.98)	−0.717 (1.46)	−0.456** (2.38)
Value added in 3-digit ISIC in a year/total value added by the country in that year × innovation intensity	−0.000 (1.07)	−0.000 (0.02)	−0.000 (0.75)	−0.000 (1.05)	−0.000 (0.00)	−0.000 (0.77)
Log of country's imports to the U.S. in the 3-digit ISIC industry in a year × innovation intensity	−0.001 (0.32)	−0.003 (0.94)	−0.000 (0.28)	−0.001 (0.30)	−0.003 (0.86)	−0.000 (0.33)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year × innovation intensity	0.004** (2.04)	0.002 (0.71)	0.004*** (2.73)	0.004** (2.10)	0.002 (0.88)	0.004*** (2.73)
Log of GDP per capita	1.075*** (3.42)	1.782*** (3.50)	1.003*** (4.97)	1.055*** (3.35)	1.790*** (3.53)	0.972*** (4.83)
Value added in 3-digit ISIC in a year/total value added by the country in that year	0.000 (1.03)	−0.000 (0.13)	0.000 (0.67)	0.000 (1.01)	−0.000 (0.14)	0.000 (0.69)
Log of country's imports to the U.S. in the 3-digit ISIC industry in a year	−0.000 (0.07)	0.003 (1.05)	0.000 (0.01)	−0.000 (0.09)	0.003 (0.97)	0.000 (0.06)
Log of country's exports to the U.S. in the 3-digit ISIC industry in a year	0.003* (1.67)	0.001 (0.49)	0.003** (2.43)	0.003* (1.73)	0.002 (0.65)	0.003** (2.42)
U.S. patent class dummies	Yes	Yes	Yes	Yes	Yes	Yes
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Application year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Country dummies × innovation intensity	Yes	Yes	Yes	Yes	Yes	Yes
Application year dummies × innovation intensity	Yes	Yes	Yes	Yes	Yes	Yes
Observations	80,440	80,440	80,440	80,440	80,440	80,440
Adjusted <i>R</i> -squared	0.60	0.62	0.63	0.60	0.62	0.63

The OLS regressions above implement the following model:

$$y_{ict} = \beta_i + [\beta_c + \beta_t + \beta_1 \delta_{ct}]^* \text{InnovationIntensity}_{i,t-1} + \beta_2 \delta_{ct} + \beta X + \varepsilon_{ict},$$

where i , c , t refer to USPTO patent class i , country c , and application year t , respectively. The sample includes patents issued by the USPTO to *non-U.S. firms* over the period 1978–2002. Innovation intensity for USPTO patent class i is measured as the median number of patents held by a *U.S. firm* in patent class i in the year t . δ_{ct} equals 1 for country c and years $t \geq m + 1$ (years $t < m + 1$) if a creditor rights reform initiated in year m increased (decreased) the rights provided to creditors; δ_{ct} equals 0 otherwise. The sample of changes in creditor rights in different countries is as displayed in Table 4. β_i , β_c , β_t denote patent class, country, and application year dummies. *Creditor Rights Change* (−2, −1) equals δ_{ct} for the years $m-2$ to $m-1$ and is zero otherwise. Similarly, *Creditor Rights Change* (0, 1) equals δ_{ct} for the years m to $m + 1$ and zero otherwise while *Creditor Rights Change* (≥ 2) equals δ_{ct} for the years $m + 2$ and beyond and zero otherwise. The numbers in parentheses denote the t -statistics computed using robust standard errors that are clustered by country. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Therefore, after controlling for the unobserved heterogeneities of various kinds through Tables 3–8, we can infer that a decrease in the creditor rights leads firms to innovate more in aggregate and, in particular, leads firms in innovation-intensive industries to innovate even more compared to firms in conservative industries.

1.5 Evidence on leverage and growth

1.5.1 Evidence from leverage choice in G-7 countries. Having confirmed the real implications of our theoretical priors, we examine next the financial implications. Specifically, we test whether an innovative firm will be financed with relatively less leverage than a conservative firm, when creditor rights become stronger. This difference-in-difference test is performed using a specification similar to that in Equation (2), with y_{ict} being replaced by leverage for a given firm in country c in year t , and the measure of innovation (patenting intensity) being employed on the right-hand side at the level of two-digit SIC industry of the firm in that country.

To be consistent with the existing literature (e.g., Rajan and Zingales 1995), we include as control variables other firm characteristics (tangible assets measured as property, plant and equipment by assets, profitability measured as EBITDA by assets, log of sales, and market-to-book ratio), as well as their interactions with country dummies. Due to limits on data availability, we focus on the G-7 countries using the *Worldscope* database over the period 1990–2005. This sample of countries has reasonably well-developed debt and equity markets. Again, we test whether the coefficient β_1 on the interaction between creditor rights and innovation intensity is negative. This is indeed what we find in Table 9. In particular, when creditor rights are stronger, innovative industries take on relatively less leverage compared to other industries. This finding stays robust to different measures of leverage (book debt, market debt, debt inclusive of all nonequity liabilities, and debt net of cash and cash equivalents), though it is statistically insignificant in the case of net market debt as the leverage proxy.

This finding is important for two reasons. First, it shows that the theoretical backdrop finds support not only in its result linking creditor rights and innovation, but also in the specific mechanism we conjecture to be at play, which is that leverage is costly as a means of financing for innovative firms when creditor rights are strong. In other words, firms in innovative industries do appear to unwind the effect of stronger creditor rights through their choice of financing. They do so by undertaking smaller quantities of debt and keeping more cash reserves in order to pursue more innovative projects. Second, this finding suggests that one ought to be cautious about the approach in the law and finance literature, which ascribes greater lending associated with stronger creditor rights (at least implicitly) as an improvement in welfare and efficiency. Our results lead one to view such claims with caution as the change in creditor rights may be associated with a change in the underlying real activity, and the reason why stronger creditor rights lead to greater lending is that they

Table 9
Tests of the leverage hypothesis using data for G-7 countries

	(1) Book debt	(2) Market debt	(3) All nonequity liabilities	(4) Net book debt	(5) Net market debt	(6) Net all nonequity liabilities
Creditor rights × innovation intensity measured at the 2-digit SIC level	−0.59%*** (2.99)	−0.43%** (2.21)	−0.71%*** (2.96)	−0.81%*** (2.79)	−0.34% (1.04)	−0.95%*** (2.75)
Creditor rights	−1.29% (0.95)	−2.34% (1.61)	−8.13%*** (3.58)	−1.89% (0.80)	−1.45% (0.59)	−8.73%*** (2.81)
Innovation intensity	0.71%** (2.27)	0.52% (1.62)	0.85%** (2.15)	0.89%* (1.79)	−0.23% (0.27)	1.07%* (1.83)
Constant	7.02%* (1.65)	20.22%*** (4.43)	47.46%*** (6.52)	−13.53%* (1.75)	−6.97% (0.85)	26.01%** (2.53)
One if French legal origin		−0.012 (0.00)		−0.169 (0.00)		−1.346 (0.00)
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Industry × country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Other characteristics × country dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	60,680	60,669	60,674	60,696	60,685	60,696
Adjusted <i>R</i> -squared	0.20	0.26	0.28	0.29	0.13	0.30

This table shows regression of six measures of leverage on creditor score, innovation intensity (number of patents of median firm in the 2-digit SIC industry in the United States), and the interaction of creditor score and patenting intensity, and various fixed effects (country, year, and industry by country) and controls (tangibility, profitability, log of sales, and market-to-book ratio), as well as the interaction of control variables with country dummies. The sample period is from 1990–2005. The numbers in parentheses denote the *t*-statistics computed using robust standard errors that are clustered by country. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

discourage innovation in favor of more standard projects that can sustain greater borrowing.

1.5.2 Growth effects. As our final inquiry, we ask how the differential impact of creditor rights on different industries, depending upon their innovation intensity, affects the growth rates of these industries. Again, we follow the Rajan and Zingales (1998) methodology and employ, as our dependent variable, the growth rate in real value added over the period 1978–1992 for each ISIC (manufacturing) industry in a country. As in Rajan and Zingales (1998), we include the interaction of external financial dependence with various measures of a country's financial development, and control for unobserved heterogeneity at the country and industry levels through the respective fixed effects. Since we found in Table 6 that the interaction of financial development measures with patenting intensity affects our measures of innovation, we include this interaction as well. We test whether the coefficient of the interaction between creditor rights and patenting intensity accounts for growth over and above these effects. These tests also complement our results using patents as a proxy for innovation since they are devoid of any biases created by employing patents, and, in particular, USPTO patents, as a proxy for innovation.

We display the results of this test in Table 10. In Columns 1–4, we include the interaction of financial development measures with external financial

Table 10
Effect of creditor rights and innovation intensity on economic growth

Which financial development measure?	(1) Accounting standards	(2) Total capitalization to GDP	(3) Domestic credit to GDP	(4) Log of private credit to GDP per capita	(5) Accounting standards	(6) Total capitalization to GDP	(7) Domestic credit to GDP	(8) Log of private credit to GDP per capita
Creditor rights × innovation intensity measured at the 3-digit ISIC level	−0.018*** (3.39)	−0.016*** (3.23)	−0.016*** (3.27)	−0.018*** (3.38)	−0.016* (1.81)	−0.022** (2.34)	−0.023** (2.32)	−0.025*** (2.65)
Creditor rights	0.022*** (3.44)	0.018*** (3.01)	0.019*** (3.07)	0.023*** (3.44)	0.020** (2.16)	0.025** (2.55)	0.025** (2.54)	0.030*** (2.85)
Innovation intensity measured at the 3-digit ISIC level	0.046*** (3.35)	0.048*** (3.26)	0.000 ⁱⁱⁱ (.)	0.047*** (3.33)	0.000 (.)	0.000 (.)	0.000 (.)	0.068** (2.43)
Financial development measure × external dependence	−0.000 (0.42)	0.007 (0.71)	0.010 (0.51)	−0.005 (0.47)				
Financial development measure × innovation intensity measured at the 3-digit ISIC level					−0.000 (0.35)	0.034 (1.41)	0.070 (1.58)	0.027 (1.29)
One if English legal origin	0.027** (1.97)	0.034** (2.55)	−0.019** (2.34)	0.027** (2.00)	0.072* (1.86)	0.076** (2.62)	−0.004 (0.37)	0.013 (0.75)
One if German legal origin	0.138*** (9.83)	0.140*** (9.77)	0.002 (0.28)	0.138*** (9.86)	0.135*** (7.93)	0.054*** (4.14)	−0.022 (1.37)	0.131*** (8.74)
One if French legal origin	0.121*** (3.34)	0.041*** (2.79)	−0.012 (0.89)	0.043*** (2.83)	0.181*** (4.68)	0.056*** (4.02)	0.007 (0.53)	0.086** (2.50)
Country dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	464	387	408	464	464	387	408	464
Adjusted <i>R</i> -squared	0.64	0.62	0.62	0.64	0.64	0.63	0.63	0.64

The dependent variable in the regressions above is the continuously compounded growth rate in real value added for the period 1978–1992 for each ISIC industry in each country. The creditor rights index for a country is from Djankov, McLiesh, and Shleifer (2007)—a higher measure indicates stronger rights provided to creditors in bankruptcy. Innovation intensity is measured at the 3-digit ISIC level as the median number of patents issued to U.S. firms in the 3-digit ISIC industry using Hall, Jaffe, and Trajtenberg (2001). The measure of external dependence for each ISIC industry is from Rajan and Zingales (1998). We use the following proxies for financial development: (1) accounting standards is an index created by Center for International Financial Analysis and Research examining and rating companies' 1990 annual reports on their inclusion or omission of 90 items from La Porta et al. (1998), (2) total capitalization to GDP is the ratio of the sum of equity market capitalization (as reported by the IFC) and domestic credit (IFS lines 32a–32f but not 32e) to GDP from Rajan and Zingales (1998), (3) domestic private credit to GDP is the ratio of domestic credit to the private sector, which is from IFS line 32d, over GDP from Rajan and Zingales (1998), and (4) log of private credit to GDP per capita is the logarithm of the ratio of domestic private credit (IFS line 32d) to the GDP per capita from La Porta et al. (1998). The control variables include the legal origin of a country, industry-fixed effects at the level of each ISIC, and country-fixed effects. Since the country-fixed effects subsume variation in the financial development measures while industry-fixed effects do the same for external financial dependence, they are excluded from the regressions. The numbers in parentheses denote the *t*-statistics computed using robust standard errors that are clustered by country. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

dependence, while in Columns 5–8 we include its interaction with patenting intensity. Across all these specifications, we find that the coefficient of the interaction between creditor rights and patenting intensity is strongly negative. In contrast, none of the interactions of financial development with external financial dependence or with patenting intensity are statistically significant. Using the specification in Column 8, we can see that the economic magnitude of the effect is also significant. If we consider two ISIC industries that differ in the number of patents applied by one patent per year, the more innovative industry grows faster at a continuously compounded growth rate of 4.3% (1.8%) for a country with a creditor rights index of 1 (2). For a country with a creditor rights index of 4, the more innovative industry grows slower at a continuously compounded growth rate of 3.2%. Therefore, the difference in growth rates over the period 1978–1992 for two adjacent innovative industries more than doubles when we go from a creditor rights index of 1 to 2 and reverses when the index is equal to 4.

2. Discussion

2.1 USPTO patents as a proxy for innovation

To compare innovation done by firms across countries, it is crucial to employ patents filed in a single jurisdiction by firms from these countries. This is because comparing domestic patents filed in the various countries would not accurately measure differences in ex post innovation or the ex ante incentives for innovation in these countries. There are several reasons for this. First, many developing countries such as India did not recognize process patents until the recent change brought about by the implementation of the Trade-Related Aspects of Intellectual Property Rights (TRIPS) agreement (Rapp and Rozek 1990). Second, until the recent implementation of the TRIPS agreement, the length of patent protection in most developing economies was lower (13–15 years) than that in the developed countries (17–20 years) (Rapp and Rozek 1990). Third, while two countries may have identical patent laws, their enforcement may differ substantially. For example, in Argentina the patent law has duration of protection for fifteen years. However, according to Rapp and Rozek (1990), the combination of high inflation and a maximum fine fixed in 1864 means that there is practically no penalty for infringement. Finally, the scope and number of patents granted by patent offices in two different countries may be quite different due to differences in patenting practices. For example, the Japanese patent system does not allow the different technical aspects of an invention to be included in the same application, which induces inventors to multiply the number of patent applications and, consequently, reduces the scope of patents.

In contrast, comparing patents granted in one jurisdiction alleviates such concerns of heterogeneity and provides standardization across patents in the

strength of patent protection, the duration of protection, the penalties for patent infringement and therefore the nature of patent enforcement, and the patenting practices followed by the jurisdiction's patent office for all firms filing in the jurisdiction, irrespective of which country the firms belong to.

Given its status as the technological leader, USPTO is the natural single jurisdiction of choice. Lall (2003) notes that "most researchers on international technological activity use U.S. patent data for two reasons. First, practically all innovators who seek to exploit their technology internationally take out patents in the USA, given its market size and technological strength. Second, the data are readily available and can be taken to an extremely detailed level." Furthermore, the United States has the most advanced patenting system in the world (Kortum and Lerner 1999),¹⁸ and most innovating firms internationally file patents in the United States (Cantwell and Hodson 1991).¹⁹ Finally, U.S. patents are a high quality indicator of international technological activity. For example, Cantwell and Anderson (1996) note that the pattern of patenting in the United States is a good indicator of technological activity in all industrialized and newly industrializing countries. Soete and Wyatt (1983) also note that although international patenting propensities remain lower than domestic patenting propensities, international patents are, on average, of higher "quality."

However, using patents filed with the USPTO introduces potential biases since it is likely that foreign firms file patents with the USPTO because they would like to sell their products in the United States. Paci, Sassu, and Usai (1997) find that the most important reasons for a firm to apply for a patent abroad are: (i) to protect goods to be exported to the countries concerned; (ii) to protect goods that may be subsequently produced in the foreign country, either directly or through a subsidiary company; (iii) to guarantee the payment of royalties from the granting of production licenses; and (iv) to favor the establishment of patent exchange agreements, as well as the exchange of know-how and other technological information. Hence, the controls we employed in our tests to control for such systematic biases for comparative advantages and bilateral trade patterns were quite important.

2.2 Related literature

As a broad research enquiry, our work is close to the literature on endogenous growth pioneered by Romer (1986), Grossman and Helpman (1991), and Aghion and Howitt (1992). This literature posits that investment in R&D and human capital is the central source of technical progress and an essential

¹⁸ According to Kortum and Lerner (1999), the key strengths of the American system include: (i) the awarding of the patent to the first to discover an innovation, rather than the first to file for an invention (as in the Japanese and European patenting systems); (ii) the principle that patent applications would not be published until they were awarded; and (iii) the broad interpretation of patent scope through the doctrine of equivalents.

¹⁹ Cantwell and Hodson (1991) found in their study of patenting practices of the world's largest firms (from the Fortune listings) that over 85% of all these firms had recorded patents in the United States.

ingredient of growth. This theory stresses the need for government and private sector institutions that nurture competition and innovation, provide incentives for individuals to be inventive, and have positive externalities and spillover effects that can permanently raise a country's long-run growth rate. By providing empirical evidence highlighting the role of bankruptcy codes in encouraging innovation, our paper isolates an important legal institution that can affect the country's growth.

Manso (2005) considers the optimal compensation scheme that motivates innovation and shows theoretically that the optimal scheme exhibits substantial tolerance (or even reward) for failure and reward for long-term success. He discusses in the paper how debtor-friendly bankruptcy codes could be considered as a way of motivating innovation. Landier (2006) considers a setting with endogenous cost of entrepreneurial failure and shows that there might be multiple equilibria, some that foster experimentation and others that promote conservatism. He shows that the bankruptcy code can resolve the multiplicity of equilibria, making countries with debtor-friendly codes more suitable for entrepreneurship and innovation. While Manso's model does not have leverage and Landier's model takes leverage as given, Acharya, Sundaram, and John (2004) focus on the effect of bankruptcy code on the leverage choice of firms, but take the real technology of the firm as given. In contrast to these papers, we empirically investigate the effect of bankruptcy codes on innovation and leverage.

The closest empirical study to ours is the one by Fan and White (2003), who examine how changes in the personal bankruptcy law in various U.S. states (after the 1978 Act) affected entrepreneurship. In bankruptcy, the owner-entrepreneur retains assets up to the exemption level but not beyond. They find that the probability of households owning businesses is 35% higher if they live in states with unlimited rather than low exemptions. In contrast, our empirical work provides evidence that corporate bankruptcy codes affect the innovative pursuits of corporations. Baumol (2001) documents that more than 80% of innovation in the United States is done by publicly traded corporations. Hence, our empirical results are potentially important additional evidence for what affects innovation in an economy.

Acharya, Amihud, and Litov (2008) have a similar empirical objective as ours, but focus instead on measures of conservatism, such as the propensity of firms in the country to engage in diversifying mergers and reducing operating risk. They find that countries with stronger creditor rights exhibit greater conservatism and that the dominant effect is from the creditor right corresponding to whether the management stays in place during bankruptcy or not, and whether there is any stay on secured creditors' rights. In less directly related work, Chhava and Roberts (2006) and Nini, Smith, and Sufi (2006) consider the effect on firm-level investments of creditor rights, captured in the form of covenants and capital expenditure restrictions that are explicitly contained in debt contracts. An important difference between these papers and ours lies

in our focus on the ex ante effects of creditor rights rather than on the ex post ones (that is, once covenants bind or distress has occurred).

3. Conclusion

Identifying government and private means to promote innovation in economies is considered an important step toward generating sustainable long-run growth rates. In this paper, we hypothesize that debtor-friendly bankruptcy codes encourage firm-level innovation by promoting continuations upon failure. Employing industry-level cross-country data, we show that innovative industries exhibit greater intensity of patent creation, patent citation, and faster growth in countries with weaker creditor rights in bankruptcy; this finding is confirmed by within-country analysis that exploits time-series changes in creditor rights.

On the one hand, these results have important policy implications for the endogenous growth literature in that legal institutions governing financial contracts can play a first-order role in fostering innovation and growth. On the other hand, they suggest an altogether different approach to thinking about the design of bankruptcy codes in a normative sense, in particular, an approach that focuses on the ex ante real investments undertaken by firms in response to bankruptcy codes rather than on the ex post efficiency of continuation outcomes when firms are in distress. In addition, the results suggest a promising line of corporate finance enquiry, which examines how multinational companies organize their innovative and standard operations. For example, do they locate their subsidiaries internationally such that innovative operations are funded under debtor-friendly bankruptcy regimes? More generally, shedding light on firm-level effects of bankruptcy codes on innovation appears to be a fruitful agenda for further research.

Notes to Tables and Figures

ⁱFinland underwent a two-point decrease in its creditor rights, while all other countries experienced a one-point change.

ⁱⁱThe results are robust to the use of two other measures of financial development: (i) total capitalization to GDP is the ratio of the sum of equity market capitalization (as reported by the IFC) and domestic credit (IFS lines 32a–32f but not 32e) to GDP from Rajan and Zingales (1998), and (ii) domestic private credit to GDP is the ratio of domestic credit to the private sector, which is from IFS line 32d, over GDP from Rajan and Zingales (1998).

ⁱⁱⁱThe variable is dropped from the regression here due to multicollinearity.

References

Acharya, V., and K. Subramanian. 2009. Bankruptcy Codes and Innovation: A Model. Online appendix available from the *Review of Financial Studies*.

- Acharya, V., R. Sundaram, and K. John. 2004. Cross-Country Variations in Capital Structures: The Role of Bankruptcy Codes. Working Paper, London Business School.
- Acharya, V., Y. Amihud, and L. Litov. 2008. Creditor Rights and Corporate Risk-Taking. Working Paper, London Business School.
- Aghion, P., and P. Howitt. 1992. A Model of Growth Through Creative Destruction. *Econometrica* 60(2): 323–52.
- Allen, F., and D. Gale. 2000. *Comparing Financial Systems*. Cambridge, MA: MIT Press.
- Armour, J., B. R. Cheffins, and D. A. Skeel, Jr. 2002. Corporate Ownership Structure and the Evolution of Bankruptcy Law: Lessons from the UK. Working Paper, Center for Business Research and Faculty of Law, University of Cambridge.
- Baumol, W. 2001. *The Free Market Innovation Machine*. Princeton, NJ: Princeton University Press.
- Bertrand, M., and S. Mulainathan. 2003. Enjoying the Quiet Life? Corporate Governance and Managerial Preferences. *Journal of Political Economy* 111:1043–75.
- Cantwell, J., and B. Andersen. 1996. A Statistical Analysis of Corporate Technology Leadership Historically. *Economics of Innovation and New Technology* 4(3):211–34.
- Cantwell, J., and C. Hodson. 1991. Global R&D and UK Competitiveness, in M. C. Casson (ed.), *Global Research Strategy and International Competitiveness*. Oxford: Basil Blackwell.
- Chhava, S., and M. Roberts. 2006. Is Financial Contracting Costly? An Empirical Analysis of Debt Covenants and Corporate Investment. Working Paper, Wharton School of Business, University of Pennsylvania.
- Cohen, W., R. Nelson, and J. Walsh. 1996. Appropriability Conditions and Why Firms Patent and Why They Do Not in the American Manufacturing Sector. Paper presented at the OECD Conference on New Indicators for the Knowledge-Based Economy.
- Djankov, S., C. McLiesh, and A. Shleifer. 2007. Private Credit in 129 Countries. *Journal of Financial Economics* 85(2):299–329.
- Erfat, R. 2000. The Transformation of the Israeli Bankruptcy System as a Reflection of Social Changes. *Journal of Transnational Law and Policy*, 10(1):39–68.
- Fan, W., and M. J. White. 2003. Personal Bankruptcy and the Level of Entrepreneurial Activity. *Journal of Law and Economics* 46:543–67.
- Gertner, R., and D. Scharfstein. 1991. A Theory of Workouts and the Effects of Reorganization Law. *Journal of Finance* 46(4):1189–222.
- Griliches, Z. 1990. Patent Statistics as Economic Indicators: A survey. *Journal of Economic Literature* 28:1661–707.
- Grossman, G., and E. Helpman. 1991. *Innovation and Growth in the Global Economy*. Cambridge, MA: MIT Press.
- Hall, B. H., A. Jaffe, and M. Trajtenberg. 2001. The NBER Patent Citations Data File: Lessons, Insights and Methodological Tools. Working Paper, NBER.
- Hall, B. H., A. Jaffe, and M. Trajtenberg. 2005. Market Value and Patent Citations. *RAND Journal of Economics* 32:101–28.
- Haselmann, R., K. Pistor, and V. Vig, 2006, How law affects lending, Working Paper, Columbia Law School. Available at http://papers.ssrn.com/sol3/papers.cfm?abstract_id=846665 (accessed April 3, 2009).
- Hyytinen, A., I. Kuosa, and T. Takalo. 2001. Law or Finance: Evidence from Finland. Working Paper, The Research Institute of the Finnish Economy.
- Kang, N., and N. Nayar. 2004. The Evolution of Corporate Bankruptcy Law in India. *Money and Finance* (Oct. 2003–Mar. 2004):37–58.

- King, R., and R. Levine. 1993. Finance and Growth: Schumpeter Might be Right. *Quarterly Journal of Economics* 108(3):717–38.
- Kondo, M. 1999. R&D Dynamics of Creating Patents in the Japanese Industry. *Research Policy* 28(6):587–600.
- Kortum, S., and J. Lerner. 1999. What Is Behind the Recent Surge in Patenting? *Research Policy* 28:1–22.
- La Porta, R., F. Lopez-de-Silanes, and A. Shleifer. 1999. Corporate Ownership Around the World. *Journal of Finance* 54(2):471–517.
- La Porta, R., F. Lopez-de-Silanes, A. Shleifer, and R. W. Vishny. 1997. Legal Determinants of External Finance. *Journal of Finance* 52:1131–50.
- La Porta, R., F. Lopez-de-Silanes, A. Shleifer, and R. W. Vishny. 1998. Law and Finance. *Journal of Political Economy* 101:678–709.
- Lall, S. 2003. Indicators of the Relative Importance of IPRs in Developing Countries. *Research Policy* 32(9):1657–80.
- Landier, A. 2006. Entrepreneurship and the Stigma of Failure. Working Paper, Stern School of Business, New York University.
- MacGarvie, M. 2006. Do Firms Learn from International Trade? *The Review of Economics and Statistics* 88(1):46–60.
- Manso, G. 2005. Motivating Innovation. Working Paper, MIT.
- Martel, J. 1994. More on the Impact of Bankruptcy Reform in Canada. Working Paper, Centre Interuniversitaire de Recherche en Analyse des Organisations.
- Nicita, A., and M. Olarreaga. 2006. Trade, Production and Protection: 1976–2004. *World Bank Economic Review* 21(1):165–71.
- Nini, G., D. C. Smith, and A. Sufi. 2006. Creditor Control Rights and Firm Investment Policy. Working Paper, University of Chicago, Graduate School of Business.
- Paci, R., A. Sassu, and S. Usai. 1997. International Patenting and National Technological Specialization. *Technovation* 17(1):25–38.
- Pakes, A., and M. Shankerman. 1984. The Rate of Obsolescence of Patents, Research Gestation Lags, and the Private Rate of Return to Research Resources, in Z. Griliches (ed.), *R&D, Patents and Productivity*. Chicago, IL: University of Chicago Press, 98–112.
- Rajan, R., and L. Zingales. 1995. What Do We Know about Capital Structure? Some Evidence from International Data. *Journal of Finance* 50(5):1421–60.
- Rajan, R., and L. Zingales. 1998. Financial Dependence and Growth. *American Economic Review* 88:559–86.
- Rapp, R. T., and R. P. Rozek. 1990. Benefits and Costs of Intellectual Property Protection in Developing Countries. *Journal of World Trade* 24:75–102.
- Romer, P. 1986. Increasing Returns and Long-Run Growth. *Journal of Political Economy* 94(5):1002–37.
- Soete, L., and S. Wyatt. 1983. The Use of Foreign Patenting as an Internationally Comparable Science and Technology Output Indicator. *Scientometrics* 5(1):31–54.